



HP407 – Evidence Review and Synthesis for Decision Making

SUMMATIVE SUBMISSION 2017/2018

CANDIDATE NUMBER:	<table><tr><td>8</td><td>4</td><td>1</td><td>2</td><td>6</td></tr></table>	8	4	1	2	6
8	4	1	2	6		
ASSESSMENT TITLE:	The Impact of Workplace Wellness Programs on Health Care Utilisation Costs and Productivity Losses: A Systematic Review					
WORD COUNT:	3125					

Please tick this box if you would be happy for your assessment to be shared with future cohorts: ☒

The Impact of Workplace Wellness Programs on Health Care Utilisation Costs and Productivity Losses: A Systematic Review

ABSTRACT

Objective: To evaluate whether workplace wellness programs (WWPs) in Europe present a positive benefit-cost ratio from both an employer and societal perspective, considering health care utilisation costs, absenteeism costs, presenteeism losses and total combined costs.

Design: We systematically reviewed 13 studies that tested different WWPs with a total number of 3925 participants completing a workplace intervention and 3184 participants being part of the control group. Four main outcomes were considered and compared across studies: health care utilisation costs, absenteeism, presenteeism and total employee costs incurred. Benefit-cost ratios (BCR) were subsequently computed or taken directly from studies, both from an employer and societal perspective. Averages and medians of the BCRs were calculated in Excel.

Data sources: Studies were searched for in Embase, MEDLINE, Scopus and Business Source Complete from January 1st, 2013 until March 1st, 2018. Google Scholar, the Cochrane Central Register of Controlled Trials and the reference lists of all included studies and other systematic reviews were scanned for any additional relevant materials.

Eligibility criteria: Any WWP assessed alongside a randomized controlled trial (RCT) or cluster RCT was included, thereby excluding quasi-experimental or observational studies. No limitations were put on the type of WWP or characteristics of participants, but Return-to-Work programs were excluded. Attention was paid to only include studies that reported a cost-benefit analysis of the workplace intervention and focused on either absenteeism, presenteeism, health care utilisation costs or all three as outcome measures. Finally, only studies conducted in the Eurozone (€) were considered.

Results: From both a societal and employer perspective, workplace wellness programs *on average* do not present a positive financial return when considering total outcome costs of employees. The average benefit-cost ratio from an employer perspective is -3.73 and from a societal perspective -7.61, which means that for every euro spent on the WWP, €3.73 and €7.61 is lost respectively. Similar negative or neutral financial returns are found when considering absenteeism, presenteeism and health care utilisation costs separately.

Conclusions: Our results are consistent with recent systematic reviews on workplace wellness programs that indicate little to negative financial returns of workplace wellness interventions. Future research should focus on standardizing the research methodology and economic evaluation of WWPs.

Systematic review PROSPERO registration: CRD42018091972

What is already known:

- Workplace wellness programs (WWPs) have become an increasingly popular intervention for employers, aiming to promote healthier lifestyles and improve employee health and well-being, lower medical costs and increase employee productivity.
- However, recent systematic reviews and studies suggest that these programs might not be as effective in terms of cost savings as they are intended to be.

What this study adds:

- This study provides new evidence to suggest that workplace wellness programs in Europe do not give positive financial returns from both an employer and societal perspective.

1. INTRODUCTION

1.1 Background information and rationale

Over the past three decades, workplace wellness programs (WWPs) have become an increasingly popular employee benefit in Europe. According to a recent 2016 CBRE report on WWPs, 74% of companies in Europe have wellness interventions in place, a figure that is expected to rise in the next five years ^[1, 2]. WWPs commonly range from single interventions such as having an on-site gym facility to multi-component programs that aim to manage disease and lifestyle and screen for health risks, including provision of self-help education materials or the use of financial incentives to encourage healthier behaviour ^[3].

The goal of these programs from an employer perspective is often threefold: (1) to promote healthier lifestyles and improve employee health and well-being, (2) lower workplace absenteeism and (3) increase employee productivity through reducing presenteeism, defined as an employee's decreased productivity and work quality due to coming to work despite being ill ^[4, 5, 6]. Furthermore, worksite health promotion interventions might not only benefit employers through reduced productivity-related losses but may also be beneficial from a societal perspective by reducing medical care costs for the government and medical insurance companies ^[1].

Over the past fifteen years, several systematic reviews on WWPs have reported positive returns with regards to employee medical care costs, reductions in costs of absenteeism and presenteeism and even indicate that WWPs can lead to higher job satisfaction ^[7, 8, 9, 10, 11]. For example, a prominent review by Baicker, Cutler & Song (2010) shows that on average in the United States, employee medical costs are reduced by approximately \$3.27 for every dollar spent on wellness programs ^[11]. These reviews, however, focus primarily on programs in the US and the majority of studies analysed were not randomized controlled trials. Furthermore, more recent evidence suggests that WWPs might not be as effective in terms of cost savings as they are intended to be. For example, a systematic review conducted by Jacobs et al. (2017) on the economic impact of WWPs in Canada found that robust evidence on the positive financial outcomes of WWPs remains sparse ^[12]. A study by Jones, Molitor & Reif (2018) on the effects of a wellness program affecting over 12,000 employees in the US further suggests that WWPs do not significantly change employees' behaviour and ultimately lead to medical cost savings ^[13].

For this reason, this study presents a systematic review on the financial impact of WWPs in Europe. A 2015 systematic review on this topic by Martinez-Lemos in Europe exists ^[14], but only includes evidence up till 2013. This review therefore partly updates and builds on this paper yet provides additional analysis on benefit-cost ratios of WWPs and more recent, in-depth evidence on the impact of these programs on absenteeism, presenteeism and health care utilisation costs, both from an employer and societal perspective.

1.2 Objectives

The objective of this review is to evaluate whether workplace wellness programs in Europe present a positive financial return in terms of benefit-cost ratios from both an employer and societal perspective, considering health care utilisation costs, absenteeism costs, presenteeism (productivity) losses and total costs combined.

2. METHODS

2.1 Identification and selection of studies

Studies were searched for in Embase, MEDLINE, Scopus and Business Source Complete from January 1st, 2013 until March 1st, 2018. As the 2015 European systematic review included studies up till 2013, January 1st 2013 was chosen as a start date for selection coverage. Furthermore, our search strategy is based on both the 2010 American ^[11] and 2015 European reviews ^[14] yet differs in terms of outcome measures. Detailed search strategies for all four databases can be found in Appendix I.

Google Scholar and the Cochrane Central Register of Controlled Trials were also checked for articles. To ensure literature saturation, the reference lists of all included studies and other systematic reviews were scanned for any additional relevant materials. The combined list of articles was imported to Mendeley, a reference management software, to ease the selection process.

As part of the selection criteria, only studies that evaluate workplace wellness interventions alongside a randomized controlled trial (RCT) or a cluster RCT were included. Quasi-experimental and observational studies, study protocols and systematic reviews were therefore excluded. No limitations were put on the type of WWP or characteristics of participants, but studies that evaluated Return-to-Work programs were taken out as these interventions are not conducted *in* the actual workplace. Attention was paid to only include studies that reported a cost-benefit analysis of the workplace intervention focused on either absenteeism, presenteeism, health care utilisation costs or all three factors as outcome measures. Finally, only studies conducted in the Eurozone were considered to ensure that all relevant costs were computed in euros (€).

After removing duplicates in Mendeley, two review authors independently screened all included articles from A to M based on title and abstract, and two other review authors independently screened articles N to Z. Full reports for all titles and abstracts that met the inclusion criteria or for those which were uncertain were obtained and further screened for final inclusion. It should be noted that the initial search strategy did *not* include search criteria based on geographical area. This was to ensure that no relevant articles would be missed. During the selection process articles were subsequently screened for Europe.

2.2 Data extraction

Similar to the alphabetical division used in the selection process, two groups of two independent reviewers extracted data to Excel in duplicate from the final included studies. To ensure consistency across reviewers, a calibration exercise was conducted prior to the start of the data extraction. The data items extracted included intervention, company and/or industry and participant characteristics and all relevant outcome measures if given in the study, including health care utilisation costs, absenteeism and presenteeism. In order to calculate benefit-cost ratios, costs of each workplace intervention were also included.

2.3 Risk of bias

Risk of bias for all included studies was assessed using the Cochrane Collaboration's tool for assessing risk of bias in RCTs ^[15]. Similar to the selection and data extraction process, two groups of two independent reviewers collected study information and assessed the studies based on (1) selection bias, (2) performance bias, (3) detection bias, (4) attrition bias, (5) reporting bias and (6) any other type of bias.

As some studies include cluster RCTs, an additional four biases related to this type of design were evaluated: (1) risk of recruitment bias, (2) risk of baseline imbalances, (3) risk of loss of full clusters and (4) risk of incorrect statistical analyses ^[16].

2.4 Qualitative and statistical analysis

A qualitative analysis of the included studies was performed, providing a descriptive overview of company, industry setting, intervention and participant characteristics of all evaluated WWP.

As part of the original study protocol as registered to Prospero ^[17], we mentioned that we were planning to do a meta-analysis on BCRs of workplace wellness interventions based on a random-effects model. However, due to the nature of the outcome measures it was not possible to conduct a formal meta-analysis in Stata. Instead, averages and medians of the benefit-cost ratios were computed in Excel, similar to the method used in the 2010 American systematic review of Baicker, Cutler & Song ^[11].

The benefit-cost ratios for health care utilisation costs, absenteeism costs, presenteeism or productivity losses and total combined costs from both an employer and societal perspective were calculated as follows ^[11, 18, 19]:

$$\text{Benefit} - \text{Cost Ratio (BCR)} = \frac{\text{Total value of the benefits}}{\text{Total value of the costs}}$$

No BCR is calculated for health care utilisation costs from an employer perspective because in most European countries, medical care costs do not fall on the employer.

Furthermore, prices were not discounted because of two reasons: (1) most studies only report costs as annual costs, even though most cost measures are calculated in different months throughout the study period. However, as it is often unclear in which specific month this has happened, discounting for a specific month becomes difficult. (2) As average yearly CPI indices are not computed for the Netherlands, Finland and Germany and differences in inflation rates over time are marginally small, it was decided to not account for inflation.

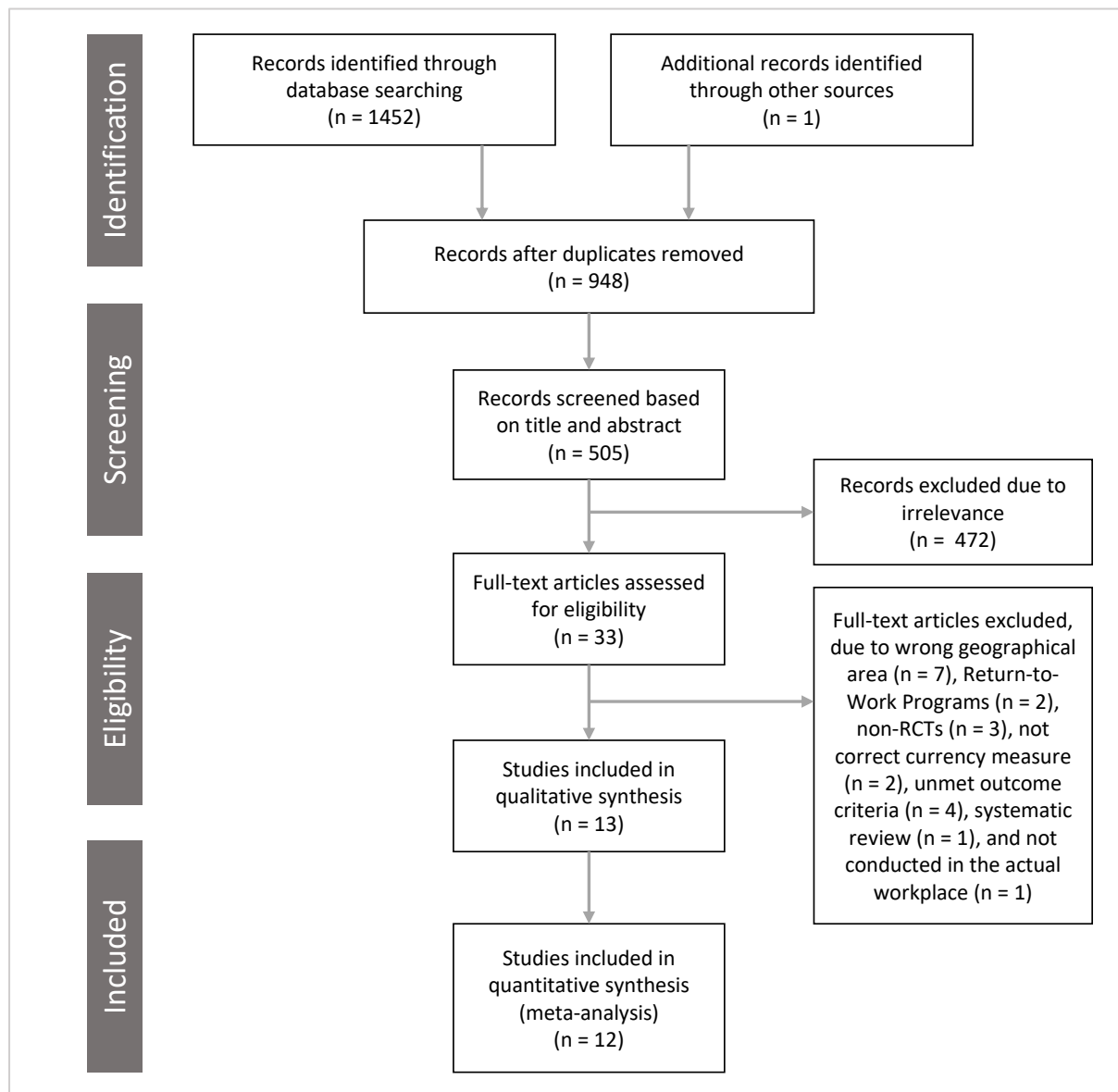
3. RESULTS

3.1 Identified and eligible studies

A total of 1452 articles were identified from the four different databases and 1 study was identified through manual search on Google Scholar. After removing duplicates, we excluded an additional 472 irrelevant articles based on title and abstract screening. The remaining 33 studies underwent full-text review, resulting in another 20 studies being excluded due to various reasons (for full reasons for exclusion please refer to Appendix II). Out of the final 13 included studies, Noben et al. (2015) was excluded from the quantitative synthesis because both presenteeism and absenteeism were calculated as ‘averted’ measures, taking into consideration costs at baseline ^[20]. However, as costs at baseline for the intervention group were already significantly higher than the control group, both outcome measures are biased and not homogeneous with the other 12 studies. An overview of the study selection process is presented in the PRISMA flow chart below ^[21].

3.2 Characteristics of included studies

A detailed description of the study characteristics can be found in Table 1. Interestingly, eleven out of thirteen studies were conducted in the Netherlands, one in Finland and one in Germany. The dominance of Dutch studies in this systematic review might explain why both the length and outcome measures of most programs are similar, as new workplace interventions in the Netherlands have to adhere to specific economic guidelines ^[22,23]. In addition to differences in country context, there are quite strong differences *between* WWPs, with no workplace intervention being exactly identical to others. However, a division can be made between interventions aimed at tackling physical versus psychological employee health problems. For example, five WWPs were implemented with the goal of improving work functioning by reducing depression, distress, anxiety or insomnia ^[19, 20, 24, 25, 26]. Furthermore, five studies analysed WWPs that tackled physical health, such as reducing body weight, managing lower back pain or rheumatoid arthritis or preventing hand eczema ^[22, 27, 28, 29, 30]. Full detailed descriptions of all evaluated workplace wellness interventions can be found in Appendix III.

Figure 1: PRISMA Flow Chart

3.3 Risk of bias within and across studies

A risk of bias assessment was completed for all 13 included studies. Most studies randomly allocated participants to either the intervention and control group and secured allocation concealment by working with an independent researcher. For this reason, all studies have relatively low risk of selection bias. However, an important characteristic of WWP evaluated alongside RCTs is that because of the nature of the intervention, blinding of participants and investigators is often not possible. It is therefore difficult to assess the extent of performance bias on the final outcome measures, which is the reason why both performance and detection bias for most studies is labelled 'unclear'. A summary overview of the risk of bias assessment is given in the table below. A further in-depth overview with relevant support of judgment is presented in Appendix IV.

Table 1: Study Characteristics

Study	Country	Workplace Intervention	Setting / Industry	Participant Population Description						Duration Intervention
				Number of Participants		Average Age		% Female		
				Intervention	Control	Intervention	Control	Intervention	Control	
Geraedts et al., 2015	Netherlands	Tackling sickness absence due to depression	Six different companies: two banking companies, two research institutes, one security company and one university	116	115	43 years	43.8 years	66.4%	58.3%	12 months
Gussenhoven et al., 2013	Netherlands	Reducing body weight	Seven different companies: two IT companies, two hospitals, an insurance company, the head office of a bank and a police force	Phone intervention: 462 Internet intervention: 464	460	Phone intervention: 43.2 years Internet intervention: 43.4 years	43.2 years	Phone intervention: 30.5% Internet intervention: 34.9%	66.5%	6 months
Noben et al., 2014	Netherlands	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	One Dutch academic medical centre	OP intervention: 207 E-Health intervention: 204	206	OP intervention: 42.56 years E-Health intervention: 37.5 years	41.83 years	OP intervention: 82.1% E-Health intervention: 83.8%	77.2%	6 months
Noben et al., 2015	Netherlands	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	One Dutch academic medical centre	207	206	42.56 years	41.83 years	82.1%	77.2%	6 months
Noben et al., 2017	Netherlands	Improving at-work productivity for workers with rheumatoid arthritis	Any workplace in the Netherlands	75	75	49.75 years	49.60 years	63	63	12 months
Oude Hengel et al., 2014	Netherlands	Prevention aimed at improving health and work ability	Six construction companies	171	122	41.8 years	44.3 years	0%	2%	6 months

Rantonen et al., 2016	Finland	Managing lower back pain	Forestry company	Combined intervention: 89 Booklet intervention: 92	83	Combined intervention: 44 years Booklet intervention: 43 years	45 years	Combined intervention: 21% Booklet intervention: 34%	24%	24 months
Thiart et al., 2016	Germany	Internet-based cognitive behavioural therapy for insomnia	Schoolteachers across the country	64	64	48.4 years	47.6 years	67.2%	81.3%	6 months
Van der Meer et al., 2015	Netherlands	Preventing hand eczema in health care workers	Several hospitals and nursing homes	876	779	40.7 years	40.8 years	80.4%	78.3%	12 months
Van Dongen et al., 2013	Netherlands	Worksite vitality intervention for older hospital workers	Two Dutch academic hospitals	367	363	52.5 years	52.3 years	74.7%	76.3%	12 months
Van Dongen et al., 2016	Netherlands	Mindfulness-based worksite intervention	Two Dutch governmental institutes	129	128	46 years	45.1 years	63.6%	71.1%	12 months
Van Dongen et al., 2017	Netherlands	Daily physical activity and relaxation intervention	24 departments of a financial service provider	Combined intervention: 92 Social intervention: 118 Physical intervention: 96	106	Combined intervention: 38 years Social intervention: 43.6 years Physical intervention: 42.2 years	40.7 years	Combined intervention: 44.6% Social intervention: 38.1% Physical intervention: 37.5%	38.7%	12 months
Van Holland et al., 2018	Netherlands	Worker's health surveillance (WHS) programme	Large Dutch meat processing company	303	683	50.6 years	<i>No personal characteristics of the control group given</i>	11.2%	<i>No personal characteristics of the control group given</i>	3 years

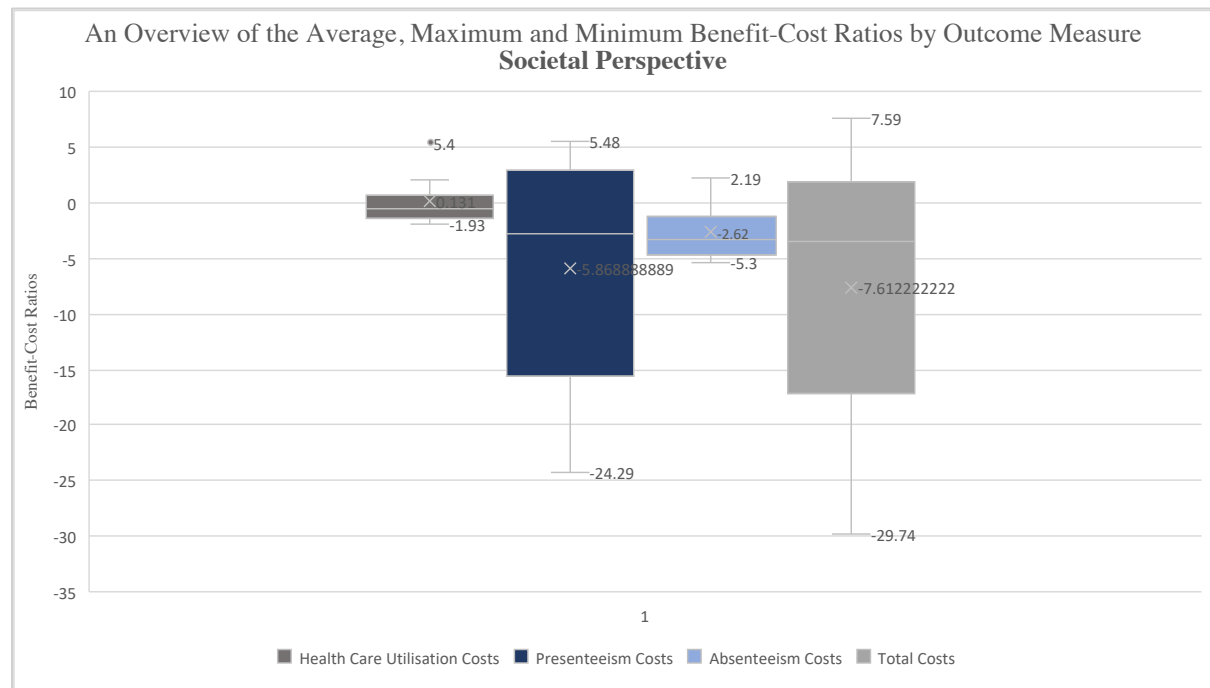
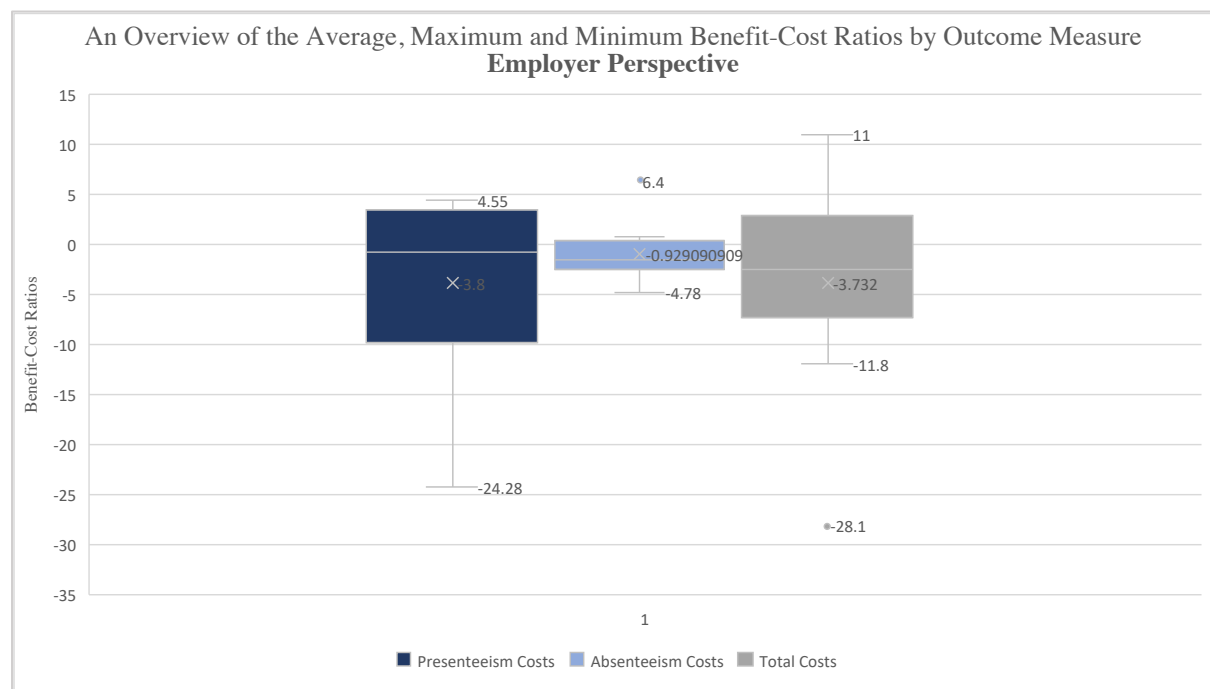
Table 2: Risk of Bias Overview

	Selection Bias Random sequence generation	Selection Bias Allocation concealment	Performance Bias Blinding of participants and personnel	Detection Bias Blinding of outcome assessment	Attrition Bias Incomplete outcome data	Reporting Bias Selective Reporting	Other Bias
Geraedts et al., 2015	Low	Low	Unclear	Unclear	High	Low	Unclear
Gussenhoven et al., 2013	Low	Low	Unclear	Low	High	Low	Unclear
Noben et al., 2014	Low	Low	Unclear	Unclear	High	Low	Unclear
Noben et al., 2015	Low	Low	Unclear	Unclear	High	Low	Low
Noben et al., 2017	Low	Low	Unclear	Unclear	Low	Low	Unclear
Oude Hengel et al., 2014	Unclear	Low	Unclear	Low	High	Low	Unclear
Rantonen et al., 2016	Low	Low	Unclear	Low	Low	Unclear	Unclear
Thiart et al., 2016	Low	Unclear	Unclear	Unclear	Low	Low	Unclear
Van der Meer et al., 2016	Low	Low	Unclear	Unclear	High	Low	Unclear
Van Dongen et al., 2013	Low	Low	Unclear	Unclear	High	Low	High
Van Dongen et al., 2016	Low	Low	Unclear	Unclear	High	Low	Unclear
Van Dongen et al., 2017	Low	Unclear	Unclear	Unclear	High	Low	Unclear
Van Holland et al., 2017	Unclear	Unclear	Unclear	Unclear	High	Low	Unclear

3.4 Results

For studies that included relevant outcome measures, BCRs were either computed manually in Excel or extracted directly from the study. BCRs for total combined costs were calculated by adding up health care utilisation, absenteeism and presenteeism costs, or were based on the total combined costs given in the study, which often also include additional costs such as sports, commuting and informal care costs.

From a societal perspective, the average financial return of WWP was negative for absenteeism, presenteeism and total combined costs, with BCRs of -2.62, -5.87 and -7.61 respectively. The average BCR for health care utilisation costs is positive yet small at 0.13. Similar results were found for BCRs calculated from an employer's perspective. A negative BCR implies that for every euro spent on the workplace intervention, either society or the employer 'loses' money. For example, an average BCR of -3.73 for total costs from an employer's perspective implies that for every euro the employer spends on the WWP, €3.73 is lost. A graphic overview of all results is presented below. Respective results tables can be found in Appendix V.

Figure 2: Overview of Results (Societal Perspective)**Figure 3: Overview of Results (Employer Perspective)**

4. DISCUSSION

4.1 Strengths and weaknesses of the study

One of the strengths of the current study is that it exclusively focuses on randomized controlled trials and provides the most up to date systematic review on the financial return of wellness programs in the European workplace. Furthermore, we adapted broad search terms as part of our search strategy as to reduce the risk of missing studies based on geographical area and to ensure that our results are generalisable for any workplace wellness intervention.

However, there are also several limitations to this study that could have had an impact on the results. First, similar to the 2010 and 2015 systematic reviews on this topic, we calculated *unweighted* averages of the benefit-cost ratios and the results therefore do not take into account any variation in intervention and study characteristics or significance of study results. Furthermore, even though 11 of the 13 included studies were conducted in the Netherlands, differences in country setting in terms of medical care and workplace environment could have acted as confounding variables. There were also large differences between studies in terms of industry setting and type of workplace intervention. Combining the results of the 13 included studies might therefore not have been appropriate and the generalisation of our results to other WWPS and European countries is limited.

4.2 Strengths and weaknesses in relation to other studies

In comparison to the meta-evaluations by Chapman (2005, 2012) ^[9, 10] and Baicker, Cutler & Song (2010) ^[11], a strength of our study is that we only included randomized controlled trials. Even though many WWPs are assessed through quasi-experimental or observational studies, by only including studies with similar methodological quality and use of guidelines for economic evaluations, we bypass any issues related to inconsistency in reported results. Furthermore, in contrast to the European review by Martinez-Lemos (2015) ^[14], this review focused solely on interventions that were actually implemented *in* the workplace, which means that return-to-work programs or interventions aimed at workers on sick leave were excluded. Our results are therefore more specific to WWPs and can be interpreted in an easier way.

In terms of weaknesses in relation to other studies, we believe that we have similar limitations to previous systematic reviews on this topic. Suggestions for future research on WWPs will therefore be further discussed in section 4.4.

4.3 Study implications

This systematic review gives evidence to suggest that workplace wellness programs in Europe are not as effective in terms of cost savings as they are intended to be. This is an important implication, especially considering that news articles continue praising wellness programs in the workplace, including a recent 2017 article on workplace health programs in the Financial

Times ^[35]. With workplace wellness programs (WWPs) becoming an increasingly popular intervention, companies should carefully consider and balance the costs and benefits of implementing a wellness program in their workplace.

4.4 Future research

Similar to limitations of previous systematic reviews, this study again illustrates the general difficulty in accurately summarizing the financial return of WWPs due to large differences in measurement methods, metrics and economic assessment. Future research on workplace interventions should therefore focus on improving the research methodology and finding a standardized approach to conducting economic evaluations of WWPs. For example, it would be useful if newly conducted RCTs could provide a comprehensive statistical analysis, so that when conducting systematic reviews on this topic, calculations of average financial returns such as BCRs can be weighted based on study significance.

5. CONCLUSIONS

This study suggests that workplace wellness programs in Europe are not cost-saving from both an employer and societal perspective. These results are consistent with other recent systematic reviews on this topic that report little to negative financial returns of workplace interventions. Future research should focus on standardizing the research methodology and economic evaluation of WWPs.

REFERENCES

- [1] CBRE. (2016). *Wellness in the Workplace: Unlocking Future Performance*. Europe: CBRE Global Workplace Solutions.
- [2] CBRE. (2016). *European Occupier Survey 2015/16*.
- [3] Mattke, S., Kapinos, K., Caloyeras, J. P., Taylor, E. A., Batorsky, B., Liu, H., . . . Newberry, S. (2015). Workplace Wellness Programs. Services Offered, Participation, and Incentives. *Rand Health Quarterly*, 5(2).
- [4] Osilla, K. C., Van Busum, K., Schnyer, C., Larkin, J. W., Eibner, C., & Mattke, S. (2012). Systematic review of the Impact of Worksite Wellness Programs. *American Journal of Management Care*.
- [5] Ammendolia, C., Côté, P., Cancelliere, C., Cassidy, J. D., Hartvigsen, J., Boyle, E., . . . Amick III, B. (2016). Healthy and productive workers: using intervention mapping to design a workplace health promotion and wellness program to improve presenteeism. *BMC Public Health*, 16(1190).
- [6] Dictionary.com. (2018, April 4). *Definition of 'presenteeism'*. Retrieved from <http://www.dictionary.com/browse/presenteeism>
- [7] Aldana, S. G. (2001, May-June). Financial impact of health promotion programs: a comprehensive review of the literature. *American Journal of Health Promotion*, 15(5), 296-320.
- [8] Parks, K. M., & Steelman, L. A. (2008, January). Organizational wellness programs: a meta-analysis. *Journal of Occupational Health Psychology*, 13(1), 58-68.
- [9] Chapman, L. S. (2005). Meta-evaluation of worksite health promotion economic return studies: 2005 update. *American Journal of Health Promotion*, 19(6), 1 10.
- [10] Chapman, L. S. (2012). Meta-Evaluation of Worksite Health Promotion Economic Return Studies: 2012 Update. *America Journal of Health Promotion*, 26, 1 12.
- [11] Baicker, K., Cutler, D., & Song, Z. (2010, February). Workplace Wellness Programs Can Generate Savings. *Health Affairs*, 29(2).
- [12] Jacobs, J. C., Yaquian, E., Burke, S. M., Rouse, M., & Zaric, G. (2017). The economic impact of workplace wellness programmes in Canada. *Occupational Medicine*, 67, 429-434.
- [13] Jones, D., Molitor, D., & Reif, J. (2018). WHAT DO WORKPLACE WELLNESS PROGRAMS DO? EVIDENCE FROM THE ILLINOIS WORKPLACE WELLNESS STUDY. *National Bureau of Economic Research* (Working Paper 24229).
- [14] Martinez-Lemos, R. I. (2015). Economic impact of corporate wellness programs in Europe: A literature review. *Journal of Occupational Health*, 57, 201-211.
- [15] Higgins, J. P., Altman, D. G., Gotzsche, P. C., Juni, P., Moher, D., Oxman, A. D., . . . Cochrane Bias Methods Group, C. S. (2011). The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. *The British Medical Journal*.
- [16] Higgins, J. P., & Green, S. (2011). 16.3.2 Assessing risk of bias in cluster-randomized trials. In *Cochrane Handbook for Systematic Reviews of Interventions* (Vol. Version 5.1.0).
- [17] Neez, E., de Wilt, E., Pentafragka, E., & Chan, A. (2018). Costs and benefits of workplace wellness programs in Europe: a systematic review - PROTOCOL. *PROSPERO*(CRD42018091972).

- [18] Investopedia.com. (2018). Retrieved from Benefit Cost-Ratio - BCR:
<https://www.investopedia.com/terms/b/bcr.asp>
- [19] Geraedts, A. S., van Dongen, J. M., Kleiboer, A. M., Wiezer, N. M., van Mechelen, W., Cuijpers, P., & Bosmans, J. E. (2015, June). Economic Evaluation of a Web-Based Guided Self-Help Intervention for Employees with Depressive Symptoms. *Journal of Occupational and Environmental Medicine*, 57(6), 666-675.
- [20] Noben, C., Evers, S., Nieuwenhuijsen, K., Ketelaar, S., Gartner, F., Sluiter, J., & Smit, F. (2015). Protecting and Promoting Mental Health of Nurses in the Hospital Setting: Is it Cost-Effective from an Employer's Perspective? *International Journal of Occupational Medicine and Environmental Health*, 28(5), 891-900.
- [21] Moher, D., Liberati, A., Tetzlaff, J., Altman, D. G., & Group, T. P. (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. *PLoS Med*, 6(7), e100097.
- [22] Noben, C., van Vilsteren, M., Boot, C., Steenbeek, R., van Schaardenburg, D., Anema, J. R., . . . de Rijk, A. (2017). Economic evaluation of an intervention program with the aim to improve at-work productivity for workers with rheumatoid arthritis. *Journal of Occupational Health*, 59, 267-279.
- [23] Krol, M., Brouwer, W., & Rutten, F. (2013, July). Productivity costs in economic evaluations: past, present, future. *Pharmacoeconomics*, 31(7), 537-549.
- [24] Noben, C., Smit, F., Nieuwenhuijsen, K., Ketelaar, S., Gartner, F., Boon, B., . . . Evers, S. (2014). Comparative cost-effectiveness of two interventions to promote work functioning by targeting mental health complaints among nurses: Pragmatic cluster randomised trial. *International Journal of Nursing Studies*, 51, 1321-1331.
- [25] Thiart, H., Ebert, D. D., Lehr, D., Nobis, S., Buntrock, C., Berking, M., . . . Riper, H. (2016). Internet-Based Cognitive Behavioral Therapy for Insomnia: A Health Economic Evaluation. *Sleep*, 39(10).
- [26] Van Dongen, J. M., van Berkel, J., Boot, C. R., Bosmans, J. E., Proper, K. I., Bongers, P. M., . . . van Wier, M. F. (2016). Long-Term Cost-Effectiveness and Return-on-Investment of a Mindfulness-Based Worksite Intervention. Results of a Randomized Controlled Trial. *Journal of Occupational and Environmental Medicine*, 58(6), 550-560.
- [27] Gussenhoven, A. H., van Wier, M. F., Bosmans, J. E., Dekkers, J. C., & van Mechelen, W. (2013). Cost-effectiveness of a distance lifestyle counselling programme among overweight employees from a company perspective, ALIFE@Work: A randomized controlled trial. *Work*, 46, 337-346.
- [28] Rantonen, J., Karppinen, J., Vehtari, A., Luoto, S., Viikari-Juntura, E., Hupli, M., . . . Taimela, S. (2016). Cost-effectiveness of providing patients with information on managing mild low-back symptoms in an occupational health setting. *BMC Public Health*, 16(316).
- [29] Van der Meer, E. W., Van Dongen, J. M., Boot, C. R., Van der Gulden, J. W., Bosmans, J. E., & Anema, J. R. (2015). Economic Evaluation of a Multifaceted Implementation Strategy for the Prevention of Hand Eczema Among Healthcare Workers in Comparison with a Control Group: The Hands4U Study. *Acta Derm Venereol*, 96, 499-504.

- [30] Van Dongen, J. M., Coffeng, J. K., van Wier, M. F., Boot, C. R., Hendriksen, I. J., van Mechelen, W., . . . van Tulder, M. W. (2017). The cost-effectiveness and return-on-investment of a combined social and physical environmental intervention in office employees. *Health Education Research*, 32(5), 384-398.
- [31] Oude Hengel, K. M., Bosmans, J. E., van Dongen, J. M., Bongers, P. M., van der Beek, A. J., & Blatter, B. M. (2014). Prevention Program at Construction Worksites Aimed at Improving Health and Work Ability Is Cost-Saving to the Employer: Results from an RCT. *American Journal of Industrial Medicine*, 57, 56-68.
- [32] Van Dongen, J. M., Strijk, J. E., Proper, K. I., van Wier, M. F., van Mechelen, W., van Tulder, M. W., & van der Beek, A. J. (2013). A Cost-effectiveness and Return-on-Investment Analysis of a Worksite Vitality Intervention Among Older Hospital Workers: Results of a Randomized Controlled Trial. *Journal of Occupational and Environmental Medicine*, 55(3), 337-346.
- [33] Van Holland, B. J., Reneman, M. F., Soer, R., Brouwer, S., & de Boer, M. R. (2018). Effectiveness and Cost-benefit Evaluation of a Comprehensive Workers' Health Surveillance Program for Sustainable Employability of Meat Processing Workers. *Journal of Occupational Rehabilitation*, 28, 107-120.
- [34] Ash, S., Murphy, H., & Chowdhury, H. (2017, September 13). How companies can intervene to transform workplace health. *Financial Times*.

Excluded Articles

- Aboagye, E., Karlsson, M. L., Hagberg, J., & Jensen, I. (2015). Cost-Effectiveness of Early Interventions for Non-Specific Low Back Pain: A Randomized Controlled Study Investigating Medical Yoga, Exercise Therapy and Self-Care Advice. *Journal of Rehabilitation Medicine*, 47, 167-173.
- Adepoju, O. E., Bolin, J. N., Ohsfeldt, R. L., Phillips, C. D., Zhao, H., Ory, M. G., & Forjuoh, S. N. (2014). Can Chronic Disease Management Programs for Patients with Type 2 Diabetes Reduce Productivity-Related Indirect Costs of the Disease? Evidence from a Randomized Controlled Trial. *Population Health Management*, 17(2).
- Arbogast, J. W., Moore-Schiltz, L., Jarvis, W. R., Harpster-Hagen, A., Hughes, J., & Parker, A. (2016, June). Impact of a Comprehensive Workplace Hand Hygiene Program on Employer Health Care Insurance Claims and Costs, Absenteeism, and Employee Perceptions and Practices. *Journal of Occupational and Environmental Medicine*, 58(6).
- Barbosa, C., Bray, J. W., Dowd, W. N., Mills, M. J., Moen, P., Wipfli, B., . . . Kelly, E. L. (2015, September). Return on Investment of a Work-Family Intervention: Evidence from the Work, Family and Health Network. *Journal of Occupational and Environmental Medicine*, 57(9), 943-951.
- Fitzgerald, S., Kirby, A., Murphy, A., Geaney, F., & Perry, I. J. (2017). A cost-analysis of complex workplace nutrition education and environmental dietary modification interventions. *BMC Public Health*, 17(49).
- Hartfiel, N., Clarke, G., Havenhand, J., Phillips, C., & Edwards, R. T. (2017). Cost-effectiveness of yoga for managing musculoskeletal conditions in the workplace. *Occupational Medicine*, 67, 687-695.

- Herman, P. M., Szczurko, O., Cooley, K., & Seely, D. (2014, February). A Naturopathic Approach to the Prevention of Cardiovascular Disease. Cost-Effectiveness Analysis of a Pragmatic Multi-Worksite Randomized Clinical Trial. *Journal of Occupational and Environmental Medicine*, 56(2).
- Kirchner, A. T., Ladd, A., Elshaw, J. J., & Schlub, J. F. (2013). An Inexpensive Workplace Initiative to Motivate High-Risk Individual Health Improvement. *Military Medicine*, 178(8), e948.
- Lehmann, B. A., Chapman, G. B., Franssen, F. M., Kok, G., & Ruiter, R. A. (2016). Changing the default to promote influenza vaccination among health care workers. *Vaccine*, 34, 1389-1392.
- Linton, S. J., Boersma, K., Traczyk, M., Shaw, W., & Nicholas, M. (2016). Early Workplace Communication and Problem Solving to Prevent Back Disability: Results of a Randomized Controlled Trial Among High-Risk Workers and Their Supervisors. *Journal of Occupational Rehabilitation*, 26, 150-159.
- Mountain, G., Windle, G., Hind, D., Walters, S., Keertharuth, A., Chatters, R., . . . Roberts, J. (2017). A preventative lifestyle intervention for older adults (lifestyle matters): a randomised controlled trial. *Age and Ageing*, 46, 627-634.
- Noben, C., Evers, S., van Genabeek, J., Nijhuis, F., & de Rijk, A. (2017). Improving a web-based employability intervention for work-disabled employees: results of a pilot economic evaluation. *Disability and Rehabilitation: Assistive Technology*, 12(3), 280-289.
- Ostbye, T., Stroo, M., Eisenstein, E. L., & Dement, J. M. (2016). The effects of two workplace weight management programs and weight loss on health care utilization and costs. *Journal of Occupational and Environmental Medicine*, 58(2), 162-169.
- Polinski, J. M., Moore, J. M., Kyrychenko, P., Gagnon, M., Matlin, O. S., Fredell, J. W., . . . Shrank, W. H. (2016, July). An Insurer's Care Transition Program Emphasizes Medication Reconciliation, Reduces Readmissions And Costs. *Health Affairs*, 35(7), 1222-1229.
- Reme, S. E., Grasdahl, A. L., Lovvik, C., Lie, S. A., & Overland, S. (2015). Work-focused cognitive-behavioural therapy and individual job support to increase work participation in common mental disorders: a randomised controlled multicentre trial. *Journal of Occupational and Environmental Medicine*, 72, 745-752.
- Sjogaard, G., Reffstrup Christensen, J., Bendix Justesen, J., Murray, M., Dalager, T., Fredslund, G. H., & Sogaard, K. (2016). Exercise is more than medicine: The working age population's well-being and productivity. *Journal of Sport and Health Science*, 5, 159-165.
- Steinberg, G., Scott, A., Honcz, J., Spettell, C., & Pradhan, S. (2015, December). Reducing Metabolic Syndrome Risk Using a Personalized Wellness Program. *Journal of Occupational and Environmental Medicine*, 57(12).
- Watson, H., Godfrey, C., McFadyen, A., McArthur, K., Stevenson, M., & Holloway, A. (2015). Screening and brief intervention delivery in the workplace to reduce alcohol-related harm: A pilot randomized controlled trial. *International Journal of Nursing Studies*, 52, 39-48.

Study Protocols of Included Studies

- Coffeng, J. K., Hendriksen, I. J., & Duijts, S. F. (2012). The development of the Be Active & Relax “Vitality in Practice” (VIP) project and design of an RCT to reduce the need for recovery in office employees. *BMC Public Health*, 12(592).
- Gärtner, F. R., Ketelaar, S. M., Smeets, O., Bolier, L., Fischer, E., van Dijk, F. J., . . . Sluiter, J. K. (2011). The Mental Vitality @ Work study: design of a randomized controlled trial on the effect of a workers' health surveillance mental module for nurses and allied health professionals. *BMC Public Health*, 11(290).
- Geraedts, A. S., Kleiboer, A. M., Wiezer, N. M., van Mechelen, W., & Cuijpers, P. (2013). Web-based guided self-help for employees with depressive symptoms (Happy@Work): design of a randomized controlled trial. *BMC Psychiatry*, 13(61).
- Oude Hengel, K. M., Joling, C. I., Proper, K. I., Blatter, B. M., & Bongers, P. M. (2010a). A worksite prevention program for construction workers: Design of a randomized controlled trial. *BMC Public Health*, doi: 10.1186/1471-2458-10-336.
- Rantonen, J., Vehtari, A., Karppinen, J., Luoto, S., Viikari-Juntura, E., Hupli, M., . . . Taimela, S. (2014). Face-to-face information combined with a booklet versus a booklet alone for treatment of mild low-back pain: a randomized controlled trial. *Scandinavian Journal of Work, Environment & Health*, 40(2), 156-166.
- Strijk, J., Proper, K., van der Beek, A., & van Mechelen, W. (2009). The Vital@Work Study. The systematic development of a lifestyle intervention to improve older workers' vitality and the design of a randomised controlled trial evaluating this intervention. *BMC Public Health*, 9(408).
- Thiart, H., Lehr, D., Ebert, D. D., Sieland, B., Berking, M., & Riper, H. (2013). Log in and breathe out: efficacy and cost-effectiveness of an online sleep training for teachers affected by work-related strain – study protocol for a randomized controlled trial. *Trials*, 14(169).
- Van Berkel, J., Proper, K. J., Boot, C. R., Bongers, P. M., & van der Beek, A. J. (2011). Mindful “Vitality in Practice”: an intervention to improve the work engagement and energy balance among workers; the development and design of the randomised controlled trial. *BMC Public Health*, 11(736).
- Van der Meer, E. W., Boot, C. R., Jungbauer, F. H., van der Klink, J. J., Rustemeyer, T., & Coenraads, P. J. (2011). Hands4U: a multifaceted strategy to implement guideline-based recommendations to prevent hand eczema in health care workers: design of a randomised controlled trial and (cost) effectiveness evaluation. *BMC Public Health*, 11(669).
- Van Vilsteren, M., Boot, C. R., Steenbeek, R., van Schaardenburg, D., Voskuyl, A. E., & Anema, J. R. (2012). An intervention program with the aim to improve and maintain work productivity for workers with rheumatoid arthritis: design of a randomized controlled trial and cost-effectiveness study. *BMC Public Health*, 12(496).
- Van Wier, M. F., Ariëns, G. A., Dekkers, J. C., Hendriksen, I. J., Pronk, N. P., Smid, T., & van Mechelen, W. (2006). ALIFE@Work: a randomised controlled trial of a distance counselling lifestyle programme for weight control among an overweight working population [ISRCTN04265725]. *BMC Public Health*, 6(140).

APPENDIX I – Search Strategies

1. Embase

1. Economic Evaluation [MeSH]
2. “economic evaluation”
3. Cost-Benefit Analysis [MeSH]
4. “return on investment”, “return-on-investment”, ROI
5. Cost control [MeSH]
6. productivity
7. saving
8. absenteeism
9. incentive*
10. spending
11. [Combine 1-10 with OR]

12. workplace
13. workplace [MeSH]
14. Worksite*
15. Corporate
16. Occupational
17. Employee
18. [Combine 12-18 with OR]

19. Health Promotion [MeSH]
20. “health promotion”
21. “lifestyle management”
22. “disease management”
23. Weight
24. Smoking
25. Exercise
26. Nutrition
27. “physical activity”
28. screening
29. wellbeing
30. wellbeing [MeSH]
31. wellness
32. prevention
33. [Combine 19-32 with OR]

34. Clinical Trial [MeSH]
35. Randomized
36. Randomized Controlled Trial [MeSH]
37. Controlled Clinical Trial [MeSH]
38. Randomly
39. RCT
40. Placebo
41. Trial

42. [Combine 34-41 with OR]
43. [Combine 11, 18, 33, 42 with AND]
44. Limit 43 to title/ abstract, English language, and year = 2013-2018

2. MEDLINE

1. Costs and Cost Analysis [MeSH]
2. "economic evaluation"
3. Cost-Benefit Analysis [MeSH]
4. Cost*
5. "return on investment", "return-on-investment", ROI
6. productivity
7. saving
8. absenteeism
9. presenteeism
10. incentive*
11. spending
12. [Combine 1-11 with OR]
13. workplace
14. workplace [MeSH]
15. Occupational Health [MeSH]
16. Occupational Health Services [MeSH]
17. Worksite*
18. Corporate
19. Occupation*
20. Employee
21. Health Benefit Plan, Employee [MeSH]
22. [Combine 13-21 with OR]
23. Health Promotion [MeSH]
24. "health promotion"
25. "lifestyle management"
26. "disease management"
27. Disease management [MeSH]
28. Weight
29. Smoking
30. Exercise
31. Nutrition
32. "physical activity"
33. screening
34. wellbeing
35. wellness
36. prevention
37. [Combine 23-36 with OR]
38. Clinical Trial [MeSH]
39. Randomized
40. Randomized Controlled Trial [MeSH]

41. Randomly
42. RCT
43. Placebo
44. Trial
45. [Combine 38-44 with OR]
46. [Combine 12, 22, 37, 45 with AND]
47. Limit 46 to title/ abstract, English language, and year = 2013-2018

3. Business Source Complete

1. Workplace [AB]
[OR]
2. Worksite [AB]
[OR]
3. Employee [AB]
[AND]
4. Incentives [AB]
[OR]
5. Productivity [AB]
[OR]
6. Absenteeism [AB]
[OR]
7. Return on investment or ROI [AB]
[OR]
8. "economic evaluation" [AB]
[AND]
9. health [AB]
[OR]
10. prevention or intervention or treatment [AB]
[OR]
11. wellness or wellbeing or well-being [AB]
[AND]
12. randomized [AB]
13. Limit to: published after 2013, language: English, Subject: Randomized Controlled Trial

3. Scopus

TITLE-ABS-KEY ("economic evaluation") OR TITLE-ABS-KEY ("return on investment") OR TITLE-ABS-KEY (roi) OR TITLE-ABS-KEY ("cost benefit analysis") OR TITLE-ABS-KEY (productivity) OR TITLE-ABS-KEY (saving*) OR TITLE-ABS-KEY ("cost control") OR TITLE-ABS-KEY (absenteeism) OR TITLE-ABS-KEY (incentive) OR TITLE-ABS-KEY (spending) AND TITLE-ABS-KEY (workplace) OR TITLE-ABS-KEY (worksite) OR TITLE-ABS-KEY (corporate) OR TITLE-ABS-KEY (employee) OR TITLE-ABS-KEY (occupational) AND TITLE-ABS-

KEY (*health* AND *promotion*) OR TITLE-ABS-KEY (*wellbeing*) OR TITLE-ABS-KEY (*wellness*) OR TITLE-ABS-KEY (*prevention*) OR TITLE-ABS-KEY ("*disease management*") OR TITLE-ABS-KEY ("*lifestyle management*") OR TITLE-ABS-KEY (*exercise*) OR TITLE-ABS-KEY (*smoking*) OR TITLE-ABS-KEY (*nutrition*) OR TITLE-ABS-KEY (*weight*) AND TITLE-ABS-KEY ("*clinical trial*") OR TITLE-ABS-KEY (*randomised*) OR TITLE-ABS-KEY (*randomly*) OR TITLE-ABS-KEY (*trial*) OR TITLE-ABS-KEY (*placebo*) OR TITLE-ABS-KEY (*rct*) AND PUBYEAR > 2013 AND (LIMIT-TO (DOCTYPE , "*ar* ") OR LIMIT-TO (DOCTYPE , "*ip* ")) AND (LIMIT-TO (LANGUAGE , "*English* "))

APPENDIX II – Reasons for Exclusion

Study	Reason for exclusion
Aboagye et al., 2015	Outcomes not measured in monetary terms.
Adepoju et al., 2014	Set in the United States.
Arbogast et al., 2016	Set in the United States.
Barbosa et al., 2015	Set in the United States.
Fitzgerald et al., 2017	Outcomes, such as productivity, were not measured in monetary terms.
Fitzgerald et al., 2018	Not a randomised controlled trial: the clusters were purposefully assigned by the study authors.
Hartfiel et al., 2016	Set in the United Kingdom with all costs measured in pounds.
Herman et al., 2014	Set in Canada.
Kirchner et al., 2013	1. Set in the United States. 2. No mention of intervention costs or outcomes expressed in monetary terms.
Lehman et al., 2016	The study only measured vaccination rates and did not report outcomes in monetary terms.
Linton et al., 2016	This study is not a proper health-economic evaluation. It does not provide outcomes in monetary terms exploitable in our review.
Mountain et al., 2017	1. Intervention not set in the workplace. 2. Outcomes not measured in monetary terms.
Noben et al., 2017	Return-To-Work study, intervention aimed at employees on sick leave or at high risk of sick leave.
Ostbye et al., 2016	Set in the United States.
Polinski et al., 2016	1. Not a randomised controlled trial. 2. Set in the United States
Reme et al., 2015	Return-To-Work study.
Sjogaard et al., 2016	Not a primary study but a systematic review of 15 studies.
Steinberg et al., 2015	1. Set in the United States. 2. Does not give program costs.
von Thile Schwartz et al., 2015	Not a randomised controlled trial.
Watson et al., 2015	1. Costs not expressed in euros (€). 2. Outcomes not expressed in monetary terms (only QALYs are considered).

APPENDIX III – Characteristics of Evaluated Workplace Wellness Programs

Study	Country	Workplace Intervention	Setting / Industry	Description Intervention
Geraedts et al., 2015	Netherlands	Tackling sickness absence due to depression	Six different companies: two banking companies, two research institutes, one security company and one university	“Happy@Work consists of 6 weekly lessons with an option of a one-week extra time in case of delay. Each lesson has a different theme related to problem solving treatment, cognitive therapy or work-related problems, but same structure: information for the theme, examples and assignments. Participants received feedback on their assignments from a coach.”
Gussenhoven et al., 2013	Netherlands	Reducing body weight	Seven different companies: two IT companies, two hospitals, an insurance company, the head office of a bank and a police force	“ALIFE@Work study: A lifestyle intervention with individual counselling through either phone or e-mail for overweight employees was developed with the aim to reduce body weight. Participants are asked to study a module and finish the assignments every two weeks, so they are able to finish the programme within six months. Participants are then contacted by their counsellor. Counselling is done according to two standardized counselling protocols (i.e., for the two communication methods): 1. Phone group: a group that received written modules with counselling by phone. 2. Internet Group: a group that received the same information through a website with counselling by email.”
Noben et al., 2014	Netherlands	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	One Dutch academic medical centre	“Two different interventions: 1. The occupational physician condition: Screening and feedback followed by referral to the occupational physician for screen-positives. 2. The e-mental health condition: Screening and feedback followed by referral and access to preventive e-mental health interventions.”
Noben et al., 2015	Netherlands	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	One Dutch academic medical centre	“The occupational physician (OP) condition: Screening and feedback followed by referral to the occupational physician for screen-positives. The subsequent OP consultation was structured according to a 7-step protocol, with the focus on identifying impairments in work functioning and providing advice on how to improve wellbeing and work functioning.”
Noben et al., 2017	Netherlands	Improving at-work productivity for workers with rheumatoid arthritis	Any workplace in the Netherlands	“The participants in the intervention group were able to take part in the intervention program consisting of two components that complement each other: 1. Integrated care: Integrated care was provided based on a case management protocol and delivered by a multidisciplinary team (n = 3), consisting of a trained clinical occupational physician (who acted as care manager), a trained occupational therapist, and the patients’ own rheumatologist. 2. Participatory workplace intervention: The participatory workplace intervention concerned workplace adaptations based on active participation and strong commitment of both the patient and, if relevant, the supervisor.”

Oude Hengel et al., 2014	Netherlands	Prevention aimed at improving health and work ability	Six construction companies	“The 6-month intervention consisted of two individual training sessions with a physical therapist to reduce physical workload, an instrument to raise awareness of the importance of rest breaks to reduce fatigue (Rest-Break tool), and two empowerment training sessions to improve the range of influence at the worksite.”
Rantonen et al., 2016	Finland	Managing lower back pain	Forestry company	“Two patient information management strategies: 1. Booklet: participants received a copy of the booklet, which information is based on the bio- psychosocial model and focuses on attitudes and inappropriate behaviour concerning LBP. The booklet has been designed to complement and support evidence-based LBP management strategies, and its key messages follow the current national guidelines. It also includes information on how to cope with LBP, avoid re-exacerbation of LBP, and emphasises that one should reassume normal activities and return to work as soon as possible. The book has been translated into Finnish from the original English version. 2. Combined: Participants in the Combined group received the booklet during their first visit, but the contents of the booklet were also reviewed to them face-to-face by the OH nurse.”
Thiart et al., 2016	Germany	Internet-based cognitive behavioural therapy for insomnia	Schoolteachers across the country	“The Internet-based CBT-I intervention (GET.ON Recovery) has been specifically tailored for stressed and sleepless employees. It mainly uses CBT-I methods such as sleep restriction, stimulus control, sleep hygiene, and cognitive interventions, supplemented by techniques effective in reducing work stress and fostering mental detachment from work-related problems derived from behavioral activation, metacognitive therapy, gratitude research, and research on boundary management. The intervention consists of six one-week modules. The participants in the intervention group received email feedback on every completed module by trained clinical psychologists (e-coaches) following a manual to ensure a standardized procedure of coaching. E-coaches also provided email reminders if participants did not complete a module within 1 week.”
Van der Meer et al., 2015	Netherlands	Preventing hand eczema in health care workers	Several hospitals and nursing homes	“The intervention group received the multifaceted implementation strategy. This strategy contained the following components: education on (the prevention of) HE, participatory working groups, role models, reminders (posters), and a leaflet setting out the evidence-based recommendations from the guideline “Contact Dermatitis” of the NVAB.”

Van Dongen et al., 2013	Netherlands	Worksite vitality intervention for older hospital workers	Two Dutch academic hospitals	“The Vital@Work intervention consisted of a Vitality Exercise Program (VEP), three Personal Vitality Coach (PVC) visits, and free fruit. Once a week, workers were invited to participate in a guided group yoga session, a guided group workout session, and 45 minutes of unsupervised vigorous physical activity (e.g. fitness and spinning). Yoga sessions were guided by qualified yoga instructors and took place at the worksite. Workout sessions were guided by certified fitness instructors and took place at a fitness centre near the worksite. PVC visits took place at the worksite.”
Van Dongen et al., 2016	Netherlands	Mindfulness-based worksite intervention	Two Dutch governmental institutes	“The intervention group participants received the 6- month Mindful VIP intervention. Briefly, the intervention consisted of mindfulness training, e- coaching, and several supporting elements (i.e., fruit and vegetables, lunch walking routes, a buddy system). The mindfulness training lasted eight weeks. Once a week, employees participated in 90- minute mindfulness group sessions (4 to 17 participants per group), given by certified trainers. Participants received hand-outs containing home- work exercises, a mindfulness exercise booklet, and an audio disc with relaxation exercises. An intranet webpage (“VIP webpage”) was developed where participants could access the intervention materials at any time. The mindfulness training was followed by an eight-week e-coaching trajectory to continue the implementation of the mindfulness principles, which was provided by the mindfulness trainers as well. If deemed necessary by the trainers, participants were referred to an occupational health service and/or an in-company health promotion activity. Throughout the six-month intervention period, free fruit and vegetables were provided at the worksite.”
Van Dongen et al., 2017	Netherlands	Daily physical activity and relaxation intervention	24 departments of a financial service provider	“Three different interventions: 1. Combined: both social and physical environmental intervention. 2. Social environment intervention: consisted of GMI, a group-based counselling technique that aims to stimulate behavioural change by exploring and resolving a participant’s ambivalence. 3. Physical environmental intervention: As part of this intervention, four ‘VIP zones’ were created at the workplace: (1) VIP Coffee Corner Zone, (2) VIP Open Office Zone, (3) VIP Meeting Zone and the (4) VIP Hall Zone. To promote stair walking, footsteps were placed on the floor of the entrance hall.”
Van Holland et al., 2018	Netherlands	Worker’s health surveillance (WHS) programme	Large Dutch meat processing company	“The POSE program was designed to identify workers who were at risk for reduced employability. Various screening tests were administered, in order to create the risk profile. For this purpose, several constructs were addressed such as physical and mental health, and physical and mental work capacity. Based on the outcomes of the screening tests, participants received recommendations on future interventions, such as a visit to the general practitioner, physiotherapist, or dietician.”

APPENDIX IV – Support for Judgment – Risk of Bias

GERAEDTS ET AL., 2015		
Domain	Support for Judgement	Review author's judgement
Selection Bias		
Random Sequence Generation	"Participants were randomized to the Web-based intervention or the CAU control group at the individual level, in blocks of 4, 6, or 8, and we used stratification at two levels: use of antidepressants and receiving treatment from a psychologist or psychiatrist at study entry." (Geraedts et al., 2015)	Low risk of bias
Allocation Concealment	"An independent researcher created the allocation schedule with a computerized random number generator, and participants were informed about their randomization outcome via e-mail" (Geraedts et al., 2015)	Low risk of bias
Performance Bias		
Blinding of participants and personnel	"Because of the nature of the intervention, blinding of participants and investigators was not possible." (Geraedts et al., 2015)	Unclear
Detection Bias		
Blinding of Outcome Assessment	"Work absence will be measured via self-report with the TiC-P. Additionally, sick leave data from the companies will be used when available and can be merged into frequency and duration of sick leave during the time of the inclusion period of the study (12 months). We will use the general work performance scale of the WHO Health and Work Performance Questionnaire (HPQ) which contains four items. Subjects will score the general work performance of other colleagues, their own usual work performance during the last two years, their overall work performance during the past four weeks, and their overall work performance during the past four weeks when compared to their colleagues. Every item will be scored from 0 to 10, with higher scores indicating better work performance. A revised version of the Trimbos and iMTA Questionnaire on Costs Associated with Psychiatric Illness (TiC-P) will be used to collect data on direct and indirect costs from the subjects." (Geraedts et al., 2013)	Unclear
Attrition Bias		
Incomplete Outcome Data	"Missing data were imputed using multiple imputation. Imputations were performed separately for the intervention and the CAU group. The imputation model included age, sex, number of working hours (these three variables were related to the "missingness" of data), and baseline and available follow-up cost and effect measure values." "Only 54% of the participants had complete data on the effect measures, and 46% on the cost measures. Even though we used multiple imputation techniques to correct for missing data, we cannot rule out the possibility of bias because of missing data. Nevertheless, there were no indications for selective attrition." (Geraedts et al., 2015) No mention of the reasons for attrition	High risk of bias
Reporting Bias		
Selective Reporting	The possibility of selective outcome reporting was examined by comparing the study protocol (Geraedts et al, 2013) with the reported results.	Low risk of bias
Other Bias	The study may be at risk of recall bias: "we only used self-report for all measures, whereas self-report may be vulnerable to recall bias."	Unclear

GUSSENHOVEN ET AL., 2013		
Domain	Support for Judgement	Review author's judgement
Selection Bias		
Random Sequence Generation	"Randomisation took place at an individual level. After baseline measurements, the employee was randomised to one of the three study groups and either to a group receiving basic weight measurements (80% of each study group) or to a group receiving additional measurements (20% of each study group). This two-step randomisation meant that there were six groups an employee could be assigned to. Randomisation to these six groups was done by block randomisation, with each block containing 15 allocations. A computerized random number generator drew up an allocation schedule. An administrative assistant put the group allocation in opaque sealed envelopes, numbered 1 to 1,500. These envelopes were taken to the locations of the baseline measurements and opened in the given order." (van Wier, 2006)	Low risk of bias
Allocation Concealment	"Randomisation was done using a concealed allocation schedule.", "The researchers were blinded for the allocation schedule" (van Wier, 2006)	Low risk of bias
Performance Bias		
Blinding of participants and personnel	"[The researchers] were not blinded for allocation after randomisation. The participants were, in consequence of the nature of the intervention, not blinded for allocation after randomisation." (van Wier, 2006); "The participants and counsellors were, in consequence of the nature of the intervention, not blinded for the intervention."	Unclear
Detection Bias		
Blinding of Outcome Assessment	"On a monthly basis the employee records full days absence from work due to sick leave. Diaries are provided at the start of each six-month measuring period. In addition, data on sickness absenteeism will be derived from employer records, from six months before baseline till two years after baseline. Work productivity is measured with the work-related questions from the WHO Health and Work Performance Questionnaire (HPQ)." (van Wier, 2006)	Low risk of bias for absenteeism, Unclear for presenteeism
Attrition Bias		
Incomplete Outcome Data	Very significant attrition: issue with both loss to follow-up at 12 months and incomplete measurements: "Participation in the study was discontinued by 370 employees." Reason provided for attrition: "The main reasons for discontinuation were lack of time or loss of interest in the study. Thirty-one participants were excluded from the analyses because of pregnancy. Two participants were excluded from the analyses because of death." "The percentage of participants with complete data differed significantly between the groups: phone group 206/462 (45%); Internet group 164/464 (35%); control group 192/460 (42%) (α 2: 8.6, $P < 0.05$). Participants with complete data had a 0.7 kg/m ² lower BMI at baseline than participants with incomplete data (28.6 vs. 29.2 kg/m ² , 95% CI 0.3; 1.0)." Statistical handling: "We used multiple imputation to handle missing data because previous studies have shown that simple procedures, like complete case analysis, can bias the estimates considerably. Nevertheless, uncertainty about the true values of the missing data is not completely resolved by this method."	High risk of bias
Reporting Bias		
Selective Reporting	The systematic reviewers evaluated the extent of selective outcome reporting by comparing the reported outcomes to the ones stated in the study protocol.	Low risk of bias
Other Bias	Because the assessment of presenteeism is based on a post-hoc questionnaire, presenteeism estimates are at risk of recall bias	Unclear

NOBEN ET AL., 2014		
Domain	Support for Judgement	Review author's judgement
Selection Bias		
Random Sequence Generation	"The randomization procedure will take place before the inclusion of the individual participants. In the randomization, we will stratify forward size. Randomization will be performed using block randomization with three departments in each block." (Gartner et al, 2011)	Low risk of bias
Allocation Concealment	"To guarantee concealment of allocation, the randomization will be performed by one researcher (KN) who is not involved in the practical recruitment of participating employers, using the computer software program Nquery Advisor." (Gartner et al, 2011)	Low risk of bias
Performance Bias		
Blinding of participants and personnel	"Researchers, managers of participating departments and Ops [occupational physicians] will not be blinded for the group allocation." (Gartner et al, 2016) Participants were blinded to information relative to other study arms: "as we have a prerandomization procedure with incomplete-double-consent design without mentioning the use of a reference group in the experimental groups and vice versa, the head (nurses) of wards and the individual employees will receive only information that is applicable to the study-arm of their wards" (Noben et al., 2014)	Unclear
Detection Bias		
Blinding of Outcome Assessment	The outcome assessment was not blinded. However, both presenteeism and absenteeism were evaluated based on participant questionnaires, this lack of blinding is leads to unclear bias of the results.	Unclear
Attrition Bias		
Incomplete Outcome Data	Frequency of drop-out: "At 6 months follow-up, dropout rates had increased to 68 (33%), 94 (45.4%), 133 (65.2%)." "The trial suffered from dropout, and our own analysis indicated that dropout had not occurred randomly" Reason for attrition: "We do not have data on reasons for non-responding to the interventions" Statistical handling of missing data: "We conducted intention-to-treat analysis using expectation maximisation imputation to handle missing data." (Noben et al., 2014)	High risk of bias
Reporting Bias		
Selective Reporting	The systematic reviewers evaluated the extent of selective outcome reporting by comparing the reported outcomes to the ones stated in the study protocol.	Low risk of bias
Other Bias - biases specific to cluster RCTs		
Risk of recruitment bias	The clusters were pre-randomized.	Low risk of bias
Risk of baseline imbalances	The study authors stated that: "There were no differences across the conditions in terms of demographics, baseline costs and work functioning. Therefore, we concluded that randomisation had resulted in a balanced trial." (Noben et al., 2014) However, they do not provide statistical analyses to substantiate this claim. Moreover, table 1 presents the baseline characteristics of the sample, and some measures seem significantly different, such as baseline medication cost or absenteeism, though the standard errors are very large.	Unclear
Risk of loss of full clusters	The study did not report who dropped out and why. Therefore, it is possible that either the drop-outs were spread over multiple clusters, or that multiple clusters were fully lost.	Unclear
Risk of incorrect statistical analyses	There is no mention that clustering was taken into account in the statistical analyses.	Unclear

NOBEN ET AL., 2015		
Domain	Support for Judgement	Review author's judgement
Selection Bias		
Random Sequence Generation	"The randomization procedure will take place before the inclusion of the individual participants. In the randomization, we will stratify forward size. Randomization will be performed using block randomization with three departments in each block." (Gartner et al, 2011)	Low risk of bias
Allocation Concealment	"To guarantee concealment of allocation, the randomization will be performed by one researcher (KN) who is not involved in the practical recruitment of participating employers, using the computer software program Nquery Advisor." (Gartner et al, 2011)	Low risk of bias
Performance Bias		
Blinding of participants and personnel	"Researchers, managers of participating departments and Ops [occupational physicians] will not be blinded for the group allocation." (Gartner et al, 2016) Participants were blinded to information relative to other study arms: "as we have a prerandomization procedure with incomplete-double-consent design without mentioning the use of a reference group in the experimental groups and vice versa, the head (nurses) of wards and the individual employees will receive only information that is applicable to the study-arm of their wards" (Noben et al., 2015)	Unclear
Detection Bias		
Blinding of Outcome Assessment	The outcome assessment was not blinded. However, both presenteeism and absenteeism were evaluated based on participant questionnaires, this lack of blinding is leads to unclear bias of the results.	Unclear
Attrition Bias		
Incomplete Outcome Data	Attrition was high (33% in the control group and 45% in the intervention group) and unbalanced. No intervention was given by study authors regarding the reason for loss to follow-up. Statistical handling of missing data: "In the main analysis, the missing data was replaced by their most likely value under the expectation maximization (EM) algorithm in the Statistical Package for the Social Sciences (SPSS) 19." "The trial data was affected by drop-out and this may have distorted the outcomes. That said, we conducted an intention-to-treat analysis by imputing missing observations under the EM algorithm. [...] Nonetheless, drop-out rates were substantial and may have biased our outcomes." (Noben et al., 2015)	High risk of bias
Reporting Bias		
Selective Reporting	The systematic reviewers evaluated the extent of selective outcome reporting by comparing the reported outcomes to the ones stated in the study protocol.	Low risk of bias
Biases: cluster RCTs		
Risk of recruitment bias	The clusters were pre-randomized.	Low risk of bias
Risk of baseline imbalances	The baseline characteristics in both groups were deemed to be similar by study authors, and table 2 supports this statement. Study authors concluded that "randomization resulted in a balanced trial". Statistical analyses would have been preferable to fully test this hypothesis.	Low risk of bias
Risk of loss of full clusters	The study did not report who dropped out and why. Therefore, it is possible that either the drop-outs were spread over multiple clusters, or that multiple clusters were fully lost.	Unclear
Risk of incorrect statistical analyses	"The analyses took into account that observations were clustered, as nurses were "nested" in different wards at the hospital. Therefore, the robust sample errors were obtained using the 1st-order Taylor series linearization within each of the 1000 bootstrap steps." (Noben et al., 2015)	Low risk of bias

NOBEN ET AL., 2017		
Domain	Support for Judgement	Review author's judgement
Selection Bias		
Random Sequence Generation	"Randomisation to either the intervention group or control group will be performed on patient level. To prevent unequal randomisation, patients will be pre-stratified by 3 prognostic factors; sex, whether the patient performs heavy physically/mentally demanding work or light physically/mentally demanding work, and the number of working hours per week. Randomisation will be performed using minimisation. Minimisation was chosen because this method makes it possible to balance groups for several prognostic factors, even in small samples." (Noben et al., 2017)	Low risk of bias
Allocation Concealment	"The result of the randomisation will be kept in a sealed envelope before handing the envelope to the participant." (Van Vilsteren et al., 2012)	Low risk of bias
Performance Bias		
Blinding of participants and personnel	"Due to the character of the intervention, patients, therapists, and researchers could not be blinded for the allocated treatment after randomization, which might increase the risk of bias." (Noben et al., 2017)	Unclear
Detection Bias		
Blinding of Outcome Assessment	"For healthcare costs, questionnaires with a six-month recall period were posted to the respondents at baseline and 6 and 12 months after randomization to measure health care utilization (i.e., health service uptake and medication usage)." "Productivity costs incorporated both absenteeism and presenteeism from work. Questionnaires with a six-month recall period were posted to the respondents at baseline and 6 and 12 months after randomization. Measuring productivity costs with the modular PROductivity and DISease Questionnaire (PRODISQ) covers all relevant aspects of the relationship between health and productivity, including absence from work, compensation mechanisms that may reduce productivity loss, and reduced productivity at work." "The Work Limitations Questionnaire was used to assess at-work productivity loss (WLQ)21." (Noben et al., 2017)	Unclear
Attrition Bias		
Incomplete Outcome Data	"Missing data were replaced by mean imputation using the mean of the outcome per group to replace the missing values per respondent. On average, 5% of the utility measurements, 4% of the WLQ measurements, and 4.8% of the cost questionnaires were missing." (Noben et al., 2017)	Low risk of bias
Reporting Bias		
Selective Reporting	The possibility of selective outcome reporting was examined by comparing the study protocol with the reported results.	Low risk of bias
Other Bias	A 6- to 12-months recall period might have led to recall bias.	Unclear

OUDE HENGEL ET AL., 2014		
Domain	Support for Judgement	Review author's judgement
Selection Bias		
Random Sequence Generation	"The randomization was performed by a research assistant who had no prior information about the departments." (Oude Hengel et al., 2010)	Unclear
Allocation Concealment	"The randomization procedure was performed by an independent research assistant, who had no prior information about the departments to ensure concealment of treatment allocation." (Oude Hengel et al., 2014)	Low risk of bias
Performance Bias		
Blinding of participants and personnel	"Because the intervention takes place at the worksite, the participants, their supervisors and the trainers cannot be blinded to the group assignment." (Oude Hengel et al., 2014)	Unclear
Detection Bias		
Blinding of Outcome Assessment	The outcome assessment was not blinded. Assessment of absenteeism: "Sickness absenteeism data were gathered from continuous registration systems of the six participating companies. Gross annual salaries of participants were estimated using the average annual salaries of construction workers, which were divided into seven age categories. Labor costs associated with 1 hr of sickness absenteeism were calculated by dividing these gross annual salaries including holiday allowances and premiums by the average number of annual working hours for construction workers. Using the Human Capital Approach [Drummond et al., 2005], absenteeism costs were estimated by multiplying the total number of sickness absenteeism hours during follow-up by the hourly labor costs." Assessment of presenteeism: "Presenteeism (i.e., reduced productivity while at work) was assessed at baseline, 3-, 6-, and 12- month follow-up using an item of the World Health Organization and Work Performance Questionnaire (WHO- HPQ)." (Oude Hengel et al., 2014) However, because the evaluation of presenteeism and absenteeism leaves little room for interpretation by the assessors, this lack of blinding is unlikely to bias the results.	Abenteeism: low risk of bias, presenteeism : unclear
Attrition Bias		
Incomplete Outcome Data	Significant drop-out, which may be non-random. No reasons for attrition given. "Workers with complete follow-up data reported significantly higher work ability at baseline than workers with incomplete follow-up data (15.9 vs. 15.3, respectively)." (Oude Hengel et al., 2014) "A second limitation concerns the high loss to follow-up on the outcomes. Complete follow-up data at 12 months (effect measures and cost data) were derived from 128 workers (44%). To account for this, missing data were imputed using the fully conditional specification and predictive mean matching, which is acknowledged as a better way to deal with missing data than complete-cases analyses." but "it could be hypothesized that the missing completely at-random assumption (i.e., missing data depend neither on the unobserved nor the observed data) was not met." (Oude Hengel et al., 2014)	High risk of bias
Reporting Bias		
Selective Reporting	The systematic reviewers evaluated the extent of selective outcome reporting by comparing the reported outcomes to the ones stated in the study protocol.	Low risk of bias
Other Bias - biases specific to cluster RCTs		
Risk of recruitment bias	N/A	Unclear
Risk of baseline imbalances	"Baseline differences between the intervention and control group were checked using the unpaired Student's t-test (continuous variables), and Pearson's chi-square test (dichotomous variables)." "At baseline, workers in the intervention group were more better educated and had slightly better mental health than workers in the control group." (Oude Hengel et al., 2014)	Unclear
Risk of loss of full clusters	The study makes no mention of who was lost to follow-up and whether full clusters were lost.	Unclear
Risk of incorrect statistical analyses	"It should be noticed that the clustering of construction workers was ignored in the analyses because the methods are not yet fully developed for economic evaluations nor does consensus exist about the best method to do so." (Oude Hengel et al., 2014)	High risk of bias

RANTONEN ET AL., 2016		
Domain	Support for Judgement	Review author's judgement
Selection Bias		
Random Sequence Generation	"An independent biostatistician prepared the stratification into two groups by using a computer-generated randomization table with a block size of 8. Based on the randomization scheme, a research assistant prepared the sealed envelopes before the start of the study containing either a referral to one of two groups: Back Book (BB) or Back Book with Advice (BB+A)." (Rantonen et al., 2014)	Low risk of bias
Allocation Concealment	"The research assistant, OH nurse and the researchers were not able to identify the participants or the group assignments before the randomization." (Rantonen et al. 2014)	Low risk of bias
Performance Bias		
Blinding of participants and personnel	"Due to the nature of the interventions, the participants and OH professionals were not blinded to the group assignment after randomization." (Rantonen et al., 2014)	Unclear
Detection Bias		
Blinding of Outcome Assessment	"About 24 months after the employee survey (i.e. baseline), participants in all study groups were asked to report their HC utilisation during the previous 12 months, i.e. 13– 24 months after the baseline. The questionnaire included the following items: the number of visits to a physician, nurse, physiotherapist or other HC professional; each representing a unit of HC utilisation." (Rantonen et al., 2014) "In order to ensure blinded analysis, all study data were entered into the data file by persons who were not related to the study." (Rantonen et al., 2016)	Low risk of bias
Attrition Bias		
Incomplete Outcome Data	"Four subjects from the BB+A and five from the BB group quit the study due to personal reasons within the first three months but gave permission to use their data. Some participants from both groups failed to return the questionnaires at all time-points because of changes at work or reasons unknown to the researchers resulting in missing data. For the missing data, the LOCF principle was followed. At the check-up visit, the activity rates in the BB+A and BB groups were 75% and 73%, respectively. The response rate was particularly high (71%)." (Rantonen et al., 2014) "In our study, about 29 % of the study visits were missing after two years. Multiple imputation attenuated the cost- effectiveness results of the Booklet group and the results of the Combined group became less apparent (see Additional file 2). However, the main conclusions of the study remained the same, whether analysed with the imputed or the complete case data (Table 2). The follow-up visit activity was similar in both intervention groups." (Rantonen et al., 2016)	Low risk of bias
Reporting Bias		
Selective Reporting	No study protocol published.	Unclear
Other Bias	Recall bias: because the study relied on self-report and retrospective logging of healthcare utilization, the study is at risk of recall bias.	Unclear

THIART ET AL., 2016		
Domain	Support for Judgement	Review author's judgement
Selection Bias		
Random Sequence Generation	"128 currently employed schoolteachers with clinically significant insomnia symptoms and relevant work-related rumination were randomly allocated to either the guided GET.ON Recovery intervention group or a waitlist control group in a ratio of 1:1, both with unrestricted access to treatment as usual. A computer program will automatically assign a 0 or 1 to each ID number, indicating the allocation to either the intervention or a waiting list. This random assignment is made in a 1:1 ratio." (Thiart et al., 2013)	Low risk of bias
Allocation Concealment	Mentioned neither in the study protocol nor in the study report.	Unclear
Performance Bias		
Blinding of participants and personnel	Mentioned neither in the study protocol nor in the study report.	Unclear
Detection Bias		
Blinding of Outcome Assessment	"Cost data were measured with the Trimbos/iMTA questionnaire for costs associated with psychiatric illness (TiC-P37), a retrospective (covering the previous three months) self-report questionnaire. [...] Participants were instructed to specify all productivity losses. Costs due to absenteeism were based on the human capital approach. In order to measure absenteeism, participants were asked how many days they had been absent from work during the past 3 mo (work loss days). To measure costs, work loss days were then multiplied to the participant's average gross daily wage based on their self-reported monthly salary. Production losses due to presenteeism were measured by asking participants to report how many days during the past 3 mo they went to work even though they were bothered by their health problems. The number of days was then multiplied by a self-reported inefficiency score, which ranged between 0 and 1 (where 0 means as efficient as when in good health and 1 means totally inefficient) to obtain workday equivalents lost to presenteeism. [...] Subsequently, based on self-reported monthly salary, their gross wages per hour were calculated and were used to calculate the costs that occurred due to presenteeism." (Thiart et al., 2016)	Unclear
Attrition Bias		
Incomplete Outcome Data	"All of the participants (n = 64) completed the first three out of 6 modules. Overall, 61 participants (95.3%) completed all six sessions." (Thiart et al., 2016)	Low risk of bias
Reporting Bias		
Selective Reporting	Not all secondary outcomes were included in the final study as compared to the study protocol. However, the primary outcome and main cost drivers were included.	Low risk of bias
Other Bias	Because all costs were based on self-reported, retrospective data, the study is at risk of recall bias. Additionally, the study may be at risk of social desirability bias since each participant self-reported their salary.	Unclear

VAN DER MEER ET AL., 2016		
Domain	Support for Judgement	Review author's judgement
Selection Bias		
Random Sequence Generation	"A total of 48 departments (n=1,649) were randomly allocated to the implementation strategy or the control group. Randomisation to the intervention group receiving the MIS or the control group takes place on the level of departments to avoid contamination between health care workers receiving the MIS or not. Prestratification was performed based on two criteria, creating four groups: workers have contact with patient (material) versus workers have no contact with patient (material); and high vs. low risk for developing hand eczema, assessed by the occupational physician. Based on the sequence of inclusion, randomisation was performed in strata of two by an independent researcher." (Van der Meer et al., 2011)	Low risk of bias
Allocation Concealment	"Workers are not informed about the outcome of the randomisation and the design of the study i.e. the existence of the two groups (control and intervention). The department managers are explicitly requested not to inform their workers about this." (Van der Meer et al., 2011)	Low risk of bias
Performance Bias		
Blinding of participants and personnel	"It is impossible to blind researchers, occupational nurses and department managers for the intervention." (Van der Meer et al., 2011)	Unclear
Detection Bias		
Blinding of Outcome Assessment	Outcome assessment was not blinded. All measures were obtained via self-reported questionnaires.	Unclear
Attrition Bias		
Incomplete Outcome Data	"Participants with complete data were statistically significantly older and reported statistically significantly less often that they had an atopic predisposition than participants without complete data. These variables were included in the imputation model." "Analyses were performed according to the intention-to-treat principle. Descriptive statistics were used to compare baseline characteristics between the intervention and control group. Missing data were multiply imputed. Imputations were performed per study group (intervention/control) using Predictive Mean Matching and Fully Conditional Specification in IBM SPSS (v20, Chicago, IL, USA)." "A first limitation of this study is the amount of missing data. For example, only 40% of the study population completed at least 4 out of the 5 cost questionnaires. Although missing data were imputed using state-of-the-art techniques, the results are probably less reliable and valid than when a complete dataset would have been obtained." (Van der Meer et al., 2016)	High risk of bias
Reporting Bias		
Selective Reporting	The systematic reviewers evaluated the extent of selective outcome reporting by comparing the reported outcomes to the ones stated in the study protocol.	Low risk of bias
Other Bias - biases specific to cluster RCTs		
Risk of recruitment bias	There was very little possibility for participants to switch clusters based on will to participate in specific study interventions.	Low risk of bias
Risk of baseline imbalances	"There were some meaningful differences between both groups for having a high level of education (control 52%; intervention 57%), performing patient-related tasks (control 81%; intervention 69%) and having HE at baseline (control 10%; intervention 7%)." (Van der Meer et al., 2016)	Unclear
Risk of loss of full clusters	Not mentioned.	Unclear
Risk of incorrect statistical analyses	"Bootstrap replications were stratified for departments, to account for the clustering of data." (Van der Meer et al., 2016)	Low risk of bias

VAN DONGEN ET AL., 2013		
Domain	Support for Judgement	Review author's judgement
Selection Bias		
Random Sequence Generation	"After baseline measurements, they [participants] were individually randomized to the intervention or control group by a research assistant using Random Allocation Software (version 1.0, May 2004, Isfahan University of Medical Sciences, Iran)." (Van Dongen et al., 2013)	Low risk of bias
Allocation Concealment	"The research assistant had no information about the workers to ensure concealment of treatment allocation." (Van Dongen et al., 2013)	Low risk of bias
Performance Bias		
Blinding of participants and personnel	"Blinding of participants and instructors was not possible." (Strijk et al., 2009)	Unclear
Detection Bias		
Blinding of Outcome Assessment	No mention of blinding of outcome assessors. "During the study period, data on other resource use (ie, health care, absenteeism, presenteeism, and sports activities) were collected on a 3-monthly basis using retrospective questionnaires."	Unclear
Attrition Bias		
Incomplete Outcome Data	"A first limitation concerns the amount of incomplete data. For 360 workers (48%), complete follow-up data were missing. This is comparable with the amount of missing data in other CEAs of worksite interventions that were conducted alongside RCTs with a follow-up of 1 year or more. Multiple imputation was used to deal with the missing data, which is acknowledged as a more appropriate way to deal with missing data than complete-case analyses. Complete-case analysis will always be inefficient, to some degree, as the sample size is reduced and it will ignore observed cost data, effect data, or both in the excluded participants. Multiple imputation, however, relies on the assumption that data are missing at random (ie, the "missingness" depends only on the observed data and not on unobserved data), an assumption that may not necessarily hold true. Therefore, the results of the present study should be interpreted with caution." (Van Dongen et al., 2013)	High risk of bias
Reporting Bias		
Selective Reporting	The systematic reviewers evaluated the extent of selective outcome reporting by comparing the reported outcomes to the ones stated in the study protocol.	Low risk of bias
Other Bias	"Another limitation may be that cost and effect data were obtained through self-reported retrospective questionnaires, which may have caused "social desirability bias," "recall bias," or both." (Van Dongen et al., 2013)	High risk of bias

VAN DONGEN ET AL., 2016		
Domain	Support for Judgement	Review author's judgement
Selection Bias		
Random Sequence Generation	"After baseline measurements, they were randomly allocated to the intervention or control group by a research assistant using a computer-generated randomization sequence in SPSS (v15.0; SPSS Inc., Chicago, IL)." (Van Dongen et al., 2016)	Low risk of bias
Allocation Concealment	"Randomisation was performed at individual level, directly after the baseline measurement (T0), using a computer-generated randomisation sequence in SPSS by an independent researcher." (Van Berkel et al., 2011) "The research assistant was blinded to group allocation." (Van Dongen et al., 2016)	Low risk of bias
Performance Bias		
Blinding of participants and personnel	"As a result of the nature of the intervention, blinding of participants and trainers was not possible." (Van Dongen et al., 2016)	Unclear
Detection Bias		
Blinding of Outcome Assessment	"Health care utilization was assessed using three-monthly retrospective questionnaires and included primary health care (eg, general practitioner, allied health professionals), secondary health care (eg, medical specialists, hospitalization), and the use of prescribed and over-the-counter medications. [...] The use of occupational health services and/or in-company health promotion activities was assessed using six-monthly retrospective questionnaires. [...] Participants' expenses on sports membership fees and sports equipment were assessed using three-monthly retrospective questionnaires. Sickness absence at baseline (ie, a one-year period before baseline) and during follow-up, as well as gross annual salaries of participants were collected from company records. Costs associated with one sick leave day were calculated per participant by dividing their gross annual salaries including holiday allowances and premiums by their total number of workable days per year. [...] Presenteeism was assessed on a three-monthly basis using an item of 'The World Health Organization Health and Work Performance Questionnaire' (WHO-HPQ)." (Van Dongen et al., 2016)	Unclear
Attrition Bias		
Incomplete Outcome Data	"Complete follow-up data were obtained from 88% of participants on the effect measures (n /4 226; 118 intervention group participants and 108 control group participants) and from 71% on the cost measures (n /4 181; 91 intervention group participants and 90 control group participants). Data on the number of provided mindfulness and e-coaching sessions were complete for all participants." (Van Dongen et al., 2016)	High risk of bias
Reporting Bias		
Selective Reporting	The systematic reviewers evaluated the extent of selective outcome reporting by comparing the reported outcomes to the ones stated in the study protocol.	Low risk of bias
Other Bias	The study may be at risk of recall bias since part of the outcomes are assessed based on self-reported, retrospectively gathered events.	Unclear

VAN DONGEN ET AL., 2017		
Domain	Support for Judgement	Review author's judgement
Selection Bias		
Random Sequence Generation	"After enrolling 19 departments in the study, subsequently, the departments were stratified to the 'physical environmental intervention' or 'no physical environmental intervention' condition. Within these strata, departments were randomized to the 'social environmental intervention' or 'no social environmental intervention' condition by means of tossing a coin. Departments will be matched pair-wise, based on the number of employees in a department, to ensure equal numbers after randomization in the intervention and control group." (Coffeng et al., 2012)	Low risk of bias
Allocation Concealment	Allocation concealment was mentioned neither in the study protocol nor in the study report.	Unclear
Performance Bias		
Blinding of participants and personnel	Participants and personnel were not blinded.	Unclear
Detection Bias		
Blinding of Outcome Assessment	No blinding of outcome assessment. "Healthcare costs: Healthcare utilization was assessed using 3-monthly questionnaires and included primary healthcare, secondary healthcare and medications. [...] Absenteeism costs: Sickness absence and salary data were collected from company records. [...] Presenteeism costs: Presenteeism was assessed on a 3-monthly basis using 'The World Health Organization Health and Work Performance Questionnaire' (WHO-HPQ), and valued using salary data." (Van Dongen et al., 2017)	Unclear
Attrition Bias		
Incomplete Outcome Data	"A total of 83 participants (20%) were lost to follow-up. Of the participants, 8% had missing sickness absence data, and 41% and 62% had one or more missing values on the effect and all other cost measure items, respectively." "Baseline characteristics of intervention and control group participants, and participants with complete and incomplete data were compared using descriptive statistics. Missing data were multiply imputed, stratified by treatment group. Using Predictive Mean Matching and Fully Conditional Specification, 15 complete data sets were created in IBM SPSS (v20)." (Van Dongen et al., 2017)	High risk of bias
Reporting Bias		
Selective Reporting	The systematic reviewers evaluated the extent of selective outcome reporting by comparing the reported outcomes to the ones stated in the study protocol. However, not all secondary outcome measures were actually used in the final study.	Low risk of bias
Other Bias	Since all effect and some cost measures were assessed using questionnaires, which may have induced 'recall bias'.	Unclear

VAN HOLLAND ET AL., 2017		
Domain	Support for Judgement	Review author's judgement
Selection Bias		
Random Sequence Generation	"Out of the 15 potentially available plants in The Netherlands, five plants fulfilled the inclusion criteria for the FLESH study. The order of inclusion in the study was randomly assigned." (Van Holland et al., 2012)	Unclear
Allocation Concealment	Mentioned neither in the study protocol nor in the study results.	Unclear
Performance Bias		
Blinding of participants and personnel	"Participants, care providers, and researchers are not blinded for the group assignment." (Van Holland et al., 2012)	Unclear
Detection Bias		
Blinding of Outcome Assessment	Outcome assessment was not blinded. "Primary outcomes: sickness absence, work ability, productivity. Sickness absence data were provided by the company. Secondary outcomes: psychosocial workload, health status, vitality. Aspects of the psychosocial workload were measured by a short form of the second version of the Copenhagen psychosocial questionnaire (COPSOQ II). Absenteeism costs. Average salaries per plant department were obtained from the company." (Van Holland et al., 2017)	Unclear
Attrition Bias		
Incomplete Outcome Data	"Only complete (sub)scales were included in the analyses. Sickness absence data were complete, as far as people were employed at the time of the study. For the other primary and secondary outcomes, on average 49.3% of the 2,121 items each was missing (N = 303)." "There were 33 (1.8%) missing values for costs related to absenteeism, caused by the fact that some workers were laid off during the trial. In addition there were 643 (36.0%) missings for costs related to presenteeism, leading to a relatively large number of missings for total costs. Therefore we first performed multiple imputations (n=20) using chained equations with predictive mean matching. For the analyses on absenteeism costs, the imputation model consisted of the absenteeism costs for every time point, the allocation at every time point, age, day salary and plant." (Van Holland et al., 2017)	High risk of bias
Reporting Bias		
Selective Reporting	The systematic reviewers evaluated the extent of selective outcome reporting by comparing the reported outcomes to the ones stated in the study protocol.	Low risk of bias
Other Bias	The study may be at risk of recall bias.	Unclear

APPENDIX V – Results

SOCIETAL PERSPECTIVE Health Care Utilisation Costs								
Study	Type of intervention	Intervention outcome costs (€)	Control outcome costs (€)	Intervention cost (€)	Control intervention cost (€)	Cost outcome difference (€)	Cost intervention difference (€)	Benefit Cost Ratio (BCR)
Geraedts et al., 2015	Sickness absence because of depression	888(NA)	769(NA)	236(NA)	0	119 (NA)	236 [NA]	-0.50
Gussenhoven et al., 2013	Reducing body weight	X	X	X	X	X	X	X
1. Noben et al., 2014 OP Intervention	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	100.71	98.95	89	25	1.76	64 [95% CI: 52 to 76]	-0.03
2. Noben et al., 2014 E-Health Intervention	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	102.76	98.95	X	X	X	X	X
Noben et al., 2015	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	X	X	X	X	X	X	X
Noben et al., 2017	Improve work productivity of workers with rheumatoid arthritis	1551.54(NA)	1748.74(NA)	91.84(NA)	0 (NA)	-189[95% CI: -768;351]	91.84 [NA]	2.06
Oude Hengel et al., 2014	Prevention aimed at improving health and work ability	X	X	X	X	X	X	X
1. Rantonen et al., 2016 Combined intervention	Managing lower back pain	195 (700)	303 (577)	20(NA)	0	-108	20	5.4
2. Rantonen et al., 2016 Booklet intervention	Managing lower back pain	Excluded	Excluded	Excluded	Excluded	Excluded	Excluded	Excluded
Thiart et al., 2016	Internet-based cognitive behavioural therapy for insomnia	X	X	X	X	X	X	X
Van der Meer et al., 2016	Preventing hand eczema in health care workers	237 (72)	88 (27)	114 (NA)	0 (NA)	149 [95% CI: 130 - 169]	114 [NA]	-1.31
Van Dongen et al., 2013	Worksite Vitality Intervention with older hospital workers	847	593	149.33(NA)	0	254 [95% CI: -246 to 670]	149,33 [NA]	-1.70
Van Dongen et al., 2016	Mindfulness-based worksite intervention	588 (126)	495 (68)	171(NA)	0	94 [95% CI: -116 to 472]	171 [NA]	-0.55
1. Van Dongen et al., 2017 Combined intervention	Evaluation of a daily physical activity and relaxation intervention in the workplace	502 (74)	602 (116)	428(NA)	0	-100	428	0.23
2. Van Dongen et al., 2017 Social intervention	Evaluation of a daily physical activity and relaxation intervention in the workplace	742 (114)	602 (116)	392(NA)	0	140	392	-0.36
3. Van Dongen et al., 2017 Physical intervention	Evaluation of a daily physical activity and relaxation intervention in the workplace	741 (183)	602 (116)	72(NA)	0	139	72	-1.93
Van Holland et al., 2017	Evaluation of Workers' Health Surveillance (WHS) programme	X	X	X	X	X	X	X
AVERAGE								0.13
MEDIAN								-0.43
BCR calculated by the review authors		BCR as included in the study						

SOCIETAL PERSPECTIVE								
Presenteeism Costs (Productivity Loss)								
Study	Type of intervention	Intervention outcome costs (€)	Control outcome costs (€)	Intervention cost (€)	Control intervention cost (€)	Cost outcome difference (€)	Cost intervention difference (€)	Benefit Cost Ratio (BCR)
Geraedts et al., 2015	Sickness absence because of depression	13980 (1434)	15259 (1421)	236 (NA)	0 (NA)	-1278 [95% CI: -5227 to 2274]	236 [NA]	5.42
Gussenhoven et al., 2013	Reducing body weight	X	X	X	X	X	X	X
1. Noben et al., 2014 OP Intervention	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	916.09	1266.78	89	25	-350.69	64 [95% CI: 52 to 76]	5.48
2. Noben et al., 2014 E-Health Intervention	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	1016.28	1266.78	X	X	X	X	X
Noben et al., 2015	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	X	X	X	X	X	X	X
Noben et al., 2017	Improve work productivity of workers with rheumatoid arthritis	2827.32(NA)	1505.44(NA)	91.84(NA)	0 (NA)	1322 [95% CI: – 138;3078]	91.84 [NA]	-14.39
Oude Hengel et al., 2014	Prevention aimed at improving health and work ability	X	X	X	X	X	X	X
1. Rantonen et al., 2016 Combined intervention	Managing lower back pain	X	X	X	X	X	X	X
2. Rantonen et al., 2016 Booklet intervention	Managing lower back pain	X	X	X	X	X	X	X
Thiart et al., 2016	Internet-based cognitive behavioural therapy for insomnia	X	X	X	X	X	X	X
Van der Meer et al., 2016	Preventing hand eczema in health care workers	3540(1023)	775(262)	113.84(NA)	0	2765	113.84	-24.29
Van Dongen et al., 2013	Worksite Vitality Intervention with older hospital workers	11580	11475	149.33	0	105	149.33	-0.70
Van Dongen et al., 2016	Mindfulness-based worksite intervention	17293 (957)	16424 (904)	171	0	869	171	-5.08
1. Van Dongen et al., 2017 Combined intervention	Evaluation of a daily physical activity and relaxation intervention in the workplace	19289 (1579)	18068 (919)	428	0	1221	428	-2.85
2. Van Dongen et al., 2017 Social intervention	Evaluation of a daily physical activity and relaxation intervention in the workplace	17880 (923)	18068 (919)	392	0	-188	392	0.48
3. Van Dongen et al., 2017 Physical intervention	Evaluation of a daily physical activity and relaxation intervention in the workplace	19284 (1276)	18068 (919)	72	0	1216	72	-16.89
Van Holland et al., 2017	Evaluation of Workers' Health Surveillance (WHS) programme	X	X	X	X	X	X	X
AVERAGE								-5.87
MEDIAN								-2.85

SOCIETAL PERSPECTIVE								
Absenteeism Costs								
Study	Type of intervention	Intervention outcome costs (€)	Control outcome costs (€)	Intervention cost (€)	Control intervention cost (€)	Cost outcome difference (€)	Cost intervention difference (€)	Benefit Cost Ratio (BCR)
Geraedts et al., 2015	Sickness absence because of depression	7220(1774)	6984(1745)	236(NA)	0(NA)	236 [95% CI: -4687 to 4932]	236 [NA]	-1
Gussenhoven et al., 2013	Reducing body weight	X	X	X	X	X	X	X
1. Noben et al., 2014 OP Intervention	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	234.1	373.95	89	25	-139.85	64 [95% CI: 52 to 76]	2.19
2. Noben et al., 2014 E-Health Intervention	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	230.03	373.95	X	X	X	X	X
Noben et al., 2015	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	X	X	X	X	X	X	X
Noben et al., 2017	Improve work productivity of workers with rheumatoid arthritis	2854.32(NA)	2367.79(NA)	91.84(NA)	0 (NA)	487 [95% CI: -859;2117]	91.84 [NA]	-5.30
Oude Hengel et al., 2014	Prevention aimed at improving health and work ability	X	X	X	X	X	X	X
1. Rantonen et al., 2016 Combined intervention	Managing lower back pain	X	X	X	X	X	X	X
2. Rantonen et al., 2016 Booklet intervention	Managing lower back pain	X	X	X	X	X	X	X
Thiart et al., 2016	Internet-based cognitive behavioural therapy for insomnia	X	X	X	X	X	X	X
Van der Meer et al., 2016	Preventing hand eczema in health care workers	3017(992)	2545(987)	113.84(NA)	0	472 [95% CI: 91;854]	113.84	-4.15
Van Dongen et al., 2013	Worksite Vitality Intervention with older hospital workers	2793	2570	149.33	0	223	149.33	-1.49
Van Dongen et al., 2016	Mindfulness-based worksite intervention	2160 (423)	1413 (214)	171	0	747	171	-4.37
1. Van Dongen et al., 2017 Combined intervention	Evaluation of a daily physical activity and relaxation intervention in the workplace	3729 (1146)	2345 (715)	428	0	1384	428	-3.23
2. Van Dongen et al., 2017 Social intervention	Evaluation of a daily physical activity and relaxation intervention in the workplace	2901 (665)	2345 (715)	392	0	556	392	-1.42
3. Van Dongen et al., 2017 Physical intervention	Evaluation of a daily physical activity and relaxation intervention in the workplace	2691 (677)	2345 (715)	72	0	346	72	-4.81
Van Holland et al., 2017	Evaluation of Workers' Health Surveillance (WHS) programme	X	X	X	X	X	X	X
AVERAGE								-2.62
MEDIAN								-3.23

SOCIETAL PERSPECTIVE								
Total Costs								
Study	Type of intervention	Intervention outcome costs (€)	Control outcome costs (€)	Intervention cost (€)	Control intervention cost (€)	Cost outcome difference (€)	Cost intervention difference (€)	Benefit Cost Ratio (BCR)
Geraedts et al., 2015	Sickness absence because of depression	22166	23115	236	0	-949 [95% CI: -5018 to 3924]	236	4.02
Gussenhoven et al., 2013	Reducing body weight	X	X	X	X	X	X	X
1. Noben et al., 2014 OP Intervention	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	1266	1752	89	25	-486	64 [95% CI: 52 to 76]	7.59
2. Noben et al., 2014 E-Health Intervention	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	1375	1752	X	X	X	X	X
Noben et al., 2015	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	X	X	X	X	X	X	X
Noben et al., 2017	Improve work productivity of workers with rheumatoid arthritis	7389.44(NA)	5750.32(NA)	91.84(NA)	0 (NA)	1638.68[NA]	91.84 [NA]	-17.84
Oude Hengel et al., 2014	Prevention aimed at improving health and work ability	X	X	X	X	X	X	X
1. Rantonen et al., 2016 Combined intervention	Managing lower back pain	X	X	X	X	X	X	X
2. Rantonen et al., 2016 Booklet intervention	Managing lower back pain	X	X	X	X	X	X	X
Thiart et al., 2016	Internet-based cognitive behavioural therapy for insomnia	X	X	X	X	X	X	X
Van der Meer et al., 2016	Preventing hand eczema in health care workers	6794(1759)	3408(1027)	113.84(NA)	0	3386 [95% CI: 2953;4055]	113.84 [NA]	-29.74
Van Dongen et al., 2013	Worksite Vitality Intervention with older hospital workers	15773	15353	149.33	0	420	149.33	-2.81
Van Dongen et al., 2016	Mindfulness-based worksite intervention	20602	18960	171	0	1642	171	-9.60
1. Van Dongen et al., 2017 Combined intervention	Evaluation of a daily physical activity and relaxation intervention in the workplace	23903	22402 (1155)	428	0	1501	428	-3.51
2. Van Dongen et al., 2017 Social intervention	Evaluation of a daily physical activity and relaxation intervention in the workplace	22449	22402 (1155)	392	0	47	392	-0.12
3. Van Dongen et al., 2017 Physical intervention	Evaluation of a daily physical activity and relaxation intervention in the workplace	23590	22402 (1155)	72	0	1188	72	-16.50
Van Holland et al., 2017	Evaluation of Workers' Health Surveillance (WHS) programme	X	X	X	X	X	X	X
AVERAGE								-7.61
MEDIAN								-3.51

EMPLOYER PERSPECTIVE Presenteeism Costs								
Study	Type of intervention	Intervention outcome costs (€)	Control outcome costs (€)	Intervention cost (€)	Control intervention cost (€)	Cost outcome difference (€)	Cost intervention difference (€)	Benefit Cost Ratio (BCR)
Geraedts et al., 2015	Sickness absence because of depression	13980(1434)	15259(1421)	285(NA)	0	-1278 [95% CI: -5227; 2274]	285 [NA]	4.48
Gussenhoven et al., 2013	Reducing body weight	X	X	X	X	X	X	X
1. Noben et al., 2014 OP intervention	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	X	X	X	X	X	X	X
2. Noben et al., 2014 E-health intervention	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	X	X	X	X	X	X	X
Noben et al., 2015	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	Excluded	Excluded	Excluded	Excluded	Excluded	Excluded	Excluded
Oude Hengel et al., 2014	Prevention aimed at improving health and work ability	Not reported	Not reported	118(NA)	0	-537 [95% CI:-1429; 315]	118 [NA]	4.55
1. Rantonen et al., 2016 Combined intervention	Managing lower back pain	X	X	X	X	X	X	X
2. Rantonen et al., 2016 Booklet intervention	Managing lower back pain	X	X	X	X	X	X	X
Thiart et al., 2016	Internet-based cognitive behavioural therapy for insomnia	1332.07 (1750.82)	1871.70 (2190.12)	200(NA)	0	-539.63 [NA]	200 [NA]	2.70
Van der Meer et al., 2016	Preventing hand eczema in health care workers	3540(1023)	775(262)	113.84(NA)	0	2764 [95% CI: 2478; 3057]	113.84 [NA]	-24.28
Van Dongen et al., 2013	Worksite Vitality Intervention with older hospital workers	11580 (408)	11475 (396)	149.33(NA)	0	106 [95% CI:-1454;1650]	149,33 [NA]	-0.71
Van Dongen et al., 2016	Mindfulness-based worksite intervention	17293 (957)	16424 (904)	464(NA)	0	869 [95% CI: 325;3930]	464 [NA]	-1.87
1. Van Dongen et al., 2017 Combined	Evaluation of a daily physical activity and relaxation intervention in the workplace	19289 (1579)	18068 (919)	466 (NA)	0	1221 [95% CI: -1028;3415]	466 [NA]	-2.62
2. Van Dongen et al., 2017 Social intervention	Evaluation of a daily physical activity and relaxation intervention in the workplace	17880 (923)	18068 (919)	430 (NA)	0	-188 [95% CI: 1864;1507]	430 [NA]	0.44
3. Van Dongen et al., 2017 Physical intervention	Evaluation of a daily physical activity and relaxation intervention in the workplace	19284 (1276)	18068 (919)	72 (NA)	0	1216 [95% CI: -638; 3102]	72 [NA]	-16.89
Van Holland et al., 2017	Evaluation of Workers' Health Surveillance (WHS) programme	X	X	X	X	X	X	X
AVERAGE								-3.80
MEDIAN								-0.71

EMPLOYER PERSPECTIVE Absenteeism Costs								
Study	Type of intervention	Intervention outcome costs (€)	Control outcome costs (€)	Intervention cost (€)	Control intervention cost (€)	Cost outcome difference (€)	Cost intervention difference (€)	Benefit Cost Ratio (BCR)
Geraedts et al., 2015	Sickness absence because of depression	8668 (3083)	8175 (2736)	285(NA)	0	492 [95% CI: -7340 ;9187]	285 [NA]	- 1.7 [95% CI: -33.1; 25.1]
Gussenhoven et al., 2013 1. Phone intervention	Reducing body weight	3317(380)	3228(435)	201(NA)	0	89 [95% CI: -1012; 1212]	201 [NA]	-0.44
Gussenhoven et al., 2013 2. Internet intervention	Reducing body weight	3098(465)	3228(435)	178(NA)	0	-130 [95% CI: -1233; 1364]	178 [NA]	0.73
Noben et al., 2014 1. Occupational Physician intervention	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	X	X	X	X	X	X	X
2. Noben et al., 2014 E-health intervention	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	X	X	X	X	X	X	X
Noben et al., 2015	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	Excluded	Excluded	Excluded	Excluded	Excluded	Excluded	Excluded
Oude Hengel et al., 2014	Prevention aimed at improving health and work ability	1727(306)	968(146)	118(NA)	0	-760 [95% CI: -1,497; -156]	118 [NA]	6.4 [NA]
1. Rantonen et al., 2016 Combined intervention	Managing lower back pain	X	X	X	X	X	X	X
2. Rantonen et al., 2016 Booklet intervention	Managing lower back pain	X	X	X	X	X	X	X
Thiart et al., 2016	Internet-based cognitive behavioural therapy for insomnia	995.40 (1650.98)	1073.40 (2573.30)	200(NA)	0	-78 [NA]	200 [NA]	0.39
Van der Meer et al., 2016	Preventing hand eczema in health care workers	3801(1256)	3527(1265)	113.84(NA)	0	544 [95% CI: 58;1028]	113.84 [NA]	-4.78
Van Dongen et al., 2013	Worksite Vitality Intervention with older hospital workers	2793(250)	2570(249)	149.33(NA)	0	223 [95% CI: -1284;1637]	149.33 [NA]	-1.49
Van Dongen et al., 2016	Mindfulness-based worksite intervention	2160 (423)	1413 (214)	464(NA)	0	746 [95% CI: -14 ;1885]	464 [NA]	-1.61
1. Van Dongen et al., 2017 Combined	Evaluation of a daily physical activity and relaxation intervention in the workplace	4410 (1507)	2520 (843)	466(NA)	0	1889 [95% CI: -803;4964]	466 [NA]	-4.05
2. Van Dongen et al., 2017 Social intervention	Evaluation of a daily physical activity and relaxation intervention in the workplace	3073 (778)	2520 (843)	430(NA)	0	553 [95% CI: -1646;2536]	430 [NA]	-1.29
3. Van Dongen et al., 2017 Physical intervention	Evaluation of a daily physical activity and relaxation intervention in the workplace	2691 (677)	2520 (843)	72(NA)	0	171 [95% CI: -1862; 1752]	72 [NA]	-2.38
Van Holland et al., 2017	Evaluation of Workers' Health Surveillance (WHS) programme	X	X	X	X	X	X	X
AVERAGE								-0.93
MEDIAN								-1.49

EMPLOYER PERSPECTIVE Total Costs								
Study	Type of intervention	Intervention outcome costs (€)	Control outcome costs (€)	Intervention cost (€)	Control intervention cost (€)	Cost outcome difference (€)	Cost intervention difference (€)	Benefit Cost Ratio (BCR)
Geraedts et al., 2015	Sickness absence because of depression	22689(3172)	23482(2314)	285(NA)	0	-793 [95% CI: -8080; 7088]	285 [NA]	2.8 [95% CI: -25.7;27.6]
Gussenhoven et al., 2013 1. Phone intervention	Reducing body weight	X	X	X	X	X	X	X
Gussenhoven et al., 2013 2. Internet intervention	Reducing body weight	X	X	X	X	X	X	X
1. Noben et al., 2014 OP intervention	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	X	X	X	X	X	X	X
2. Noben et al., 2014 E-health intervention	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	X	X	X	X	X	X	X
Noben et al., 2015	Reducing incidents of distress, anxiety and depression and promote work functioning among nurses	Excluded	Excluded	Excluded	Excluded	Excluded	Excluded	Excluded
Oude Hengel et al., 2014	Prevention aimed at improving health and work ability	NA	NA	118(NA)	0	-1297 [CI:-1470; -1095]	118 [NA]	11 [NA]
1. Rantonen et al., 2016 Combined intervention	Managing lower back pain	X	X	X	X	X	X	X
2. Rantonen et al., 2016 Booklet intervention	Managing lower back pain	X	X	X	X	X	X	X
Thiart et al., 2016	Internet-based cognitive behavioural therapy for insomnia	2327.47(2,352.08)	2945.1(3,161.10)	200(NA)	0	-617.63 [NA]	200 [NA]	3.1 [95% CI: -1.97;8.44]
Van der Meer et al., 2016	Preventing hand eczema in health care workers	7340(1926)	4032(1271)	113.84(NA)	0	3318 [95% CI: 2800;2452]	113.84 [NA]	-28.1 [95% CI: -34.0; -21.7]
Van Dongen et al., 2013	Worksite Vitality Intervention with older hospital workers	14926(NA)	14759(NA)	149.33(NA)	0	167	149.33 [NA]	-2.21 [NA]
Van Dongen et al., 2016	Mindfulness-based worksite intervention	19565(1012)	17992(950)	464(NA)	0	1573 [95% CI: -548;4752]	464 [NA]	-2.51 [95% CI: -8.19;3.10]
1. Van Dongen et al., 2017 Combined	Evaluation of a daily physical activity and relaxation intervention in the workplace	23698(1783)	20588 (1189)	466(NA)	0	3110 [95% CI: 510;6699]	466 [NA]	-5.7 [95% CI: -11.7;0.8]
2. Van Dongen et al., 2017 Social intervention	Evaluation of a daily physical activity and relaxation intervention in the workplace	20954(1141)	20588 (1189)	430(NA)	0	366 [95% CI: -1715;3179]	430 [NA]	0.0 [95% CI: -5.3;5.2]
3. Van Dongen et al., 2017 Physical intervention	Evaluation of a daily physical activity and relaxation intervention in the workplace	21975(1388)	20588 (1189)	72(NA)	0	1387 [95% CI: -1056; 3830]	72 [NA]	-11.8 [95% CI: -65;35]
Van Holland et al., 2017	Evaluation of Workers' Health Surveillance (WHS) programme	N/A	N/A	200(NA)	0	775 [95% CI: -1077;-440]	200 [NA]	-3.9 [95% CI: -5.7;-2.5]
AVERAGE								-3.73
MEDIAN								-2.36

